

Machine Learning

Prof. Dr.-Ing. Maria Schweigel

organizational things

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The Module is divided into a theoretical and a practical part.

In the first part of the semester lessons will give you theoretical background.

In the second part of the semester the practical part starts. Therefore have a look at the practical training guide and solve the tasks. We work with the open source SW TensorFlow which has to be installed on your computer. We use our meetings to discuss about problems and results of your tasks. You have the possibility to work on your own ML project. With a presentation of your project you can earn some extra points for the written exam. Please consider that each student has to work on an own project.

Course consists of :

- Theoretical basics: Lectures and Exercises
- Practical part: examples and ML-project

Materials on study:

- Slides of lesson
- Practical training guide for practical part

Meetings:

- First part of semester theoretical basics
- Second part of semester for consultation and discussion about practical part
- presentations of students' projects at the end of semester

Exam:

- Exam preparation at one of the last meetings
- written exam at end of semester about 120 min

In libray available:

- Machine Learning Algorithms : Popular Algorithms for Data Science and Machine Learning, Bonaccorso, Giuseppe
- Machine learning for dummies / by John Paul Mueller and Luca Massaron
- Fundamentals of machine learning for predictive data analytics : algorithms, worked examples, and case studies / John D. Kelleher, Brian Mac Namee, Aoife D'Arcy
- Hands-On Neural Networks with TensorFlow 2. 0 : Understand TensorFlow, from Static Graph to Eager Execution, and Design Neural Networks, Galeone, Paolo
- Hands-On Ensemble Learning with Python : Build Highly Optimized Ensemble Machine Learning Models Using Scikit-Learn and Keras, Kyriakides, George
- Maschinelles Lernen , Alpaydın, Ethem
- Machine Learning kompakt : alles, was Sie wissen müssen / Andriy Burkov ; Übersetzung aus dem Amerikanischen von Knut Lorenzen

Ebooks:

- An Introduction to Machine Learning, Miroslav Kubat, © Springer International Publishing, ISBN 978-3-319-63912-3 ISBN 978-3-319-63913-0 (eBook)

1. Introduction to Machine Learning
History of ML, Terms of ML, Learning process, Applications of ML, Key problems of ML
2. Algorithms used for ML
Decision Trees, Probabilistic Networks (Bayesian), Nearest-Neighbor Classifiers, Neural network (NN), Support Vector Machine (SVM)
3. Methods of Machine Learning
Supervised learning, Unsupervised learning, Reinforcement learning
4. Data Science and ML
Needed data sets, Examples of data sets, Problems with data

5. Quality values of ML-Algorithms
error rate and accuracy, Confusion matrix for 2-class problems, other two class quality values, Confusion matrix for m-class problems, Mean Square Error
6. Parameter optimization
gradient method, Backpropagation, Over- and underfitting, Grid searching, Genetic algorithms
7. Tensor Flow for ML
What is TensorFlow?, Keras - The high-level API for TensorFlow, Support Vector Machine in Tensorflow

Introduction to Machine learning

First ML example

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Is the matarmelon ripe?



Many definitions exist! One possible:

“Machine learning is the technique that improves system performance by learning from experience via computational methods.”

First ML example

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4 samples of our ML example:

A toy watermelon data set

ID	color	root	sound	ripe
1	green	curly	muffled	true
2	dark	curly	muffled	true
3	green	straight	crisp	false
4	dark	slightly curly	dull	false

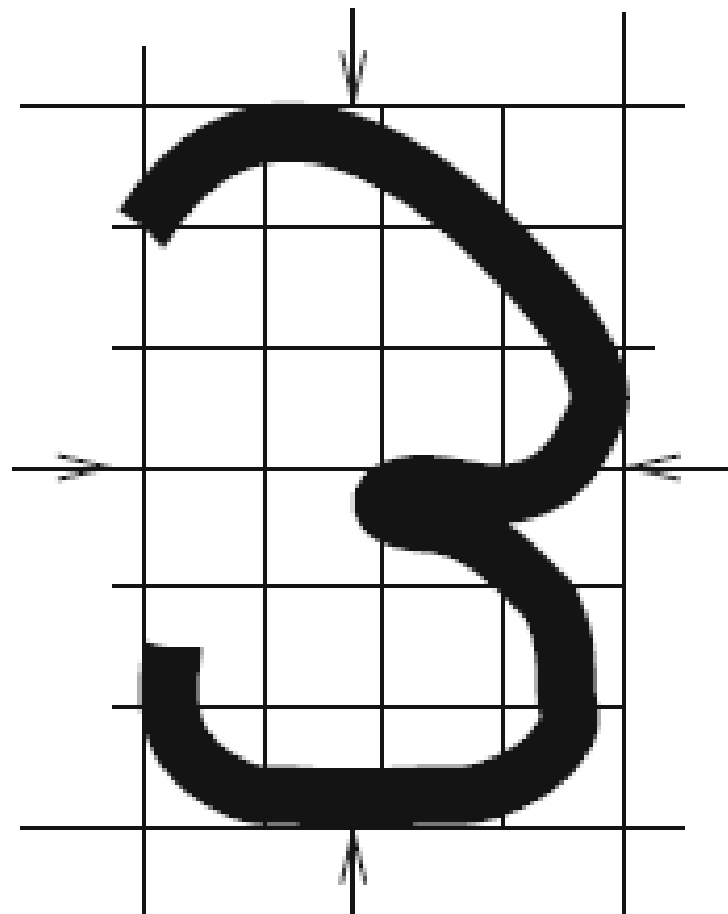


Zhou, Machine Learning, 2021

Example Character recognition

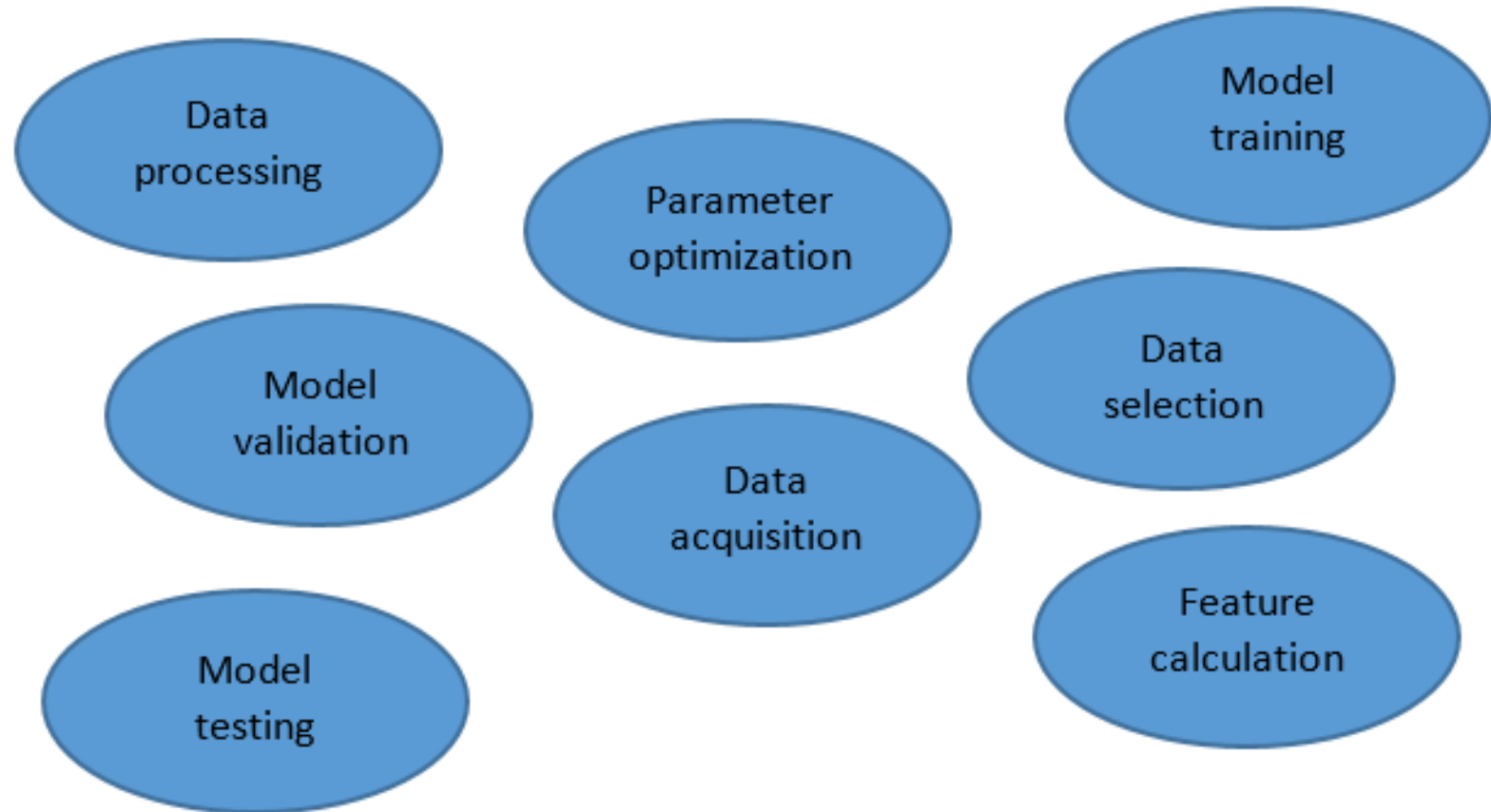
10

From 6x4 images with handwritten character
(number or letter)



Kubat, An Introduction to Machine Learning, 2017

Find correct process flow by sort single steps!



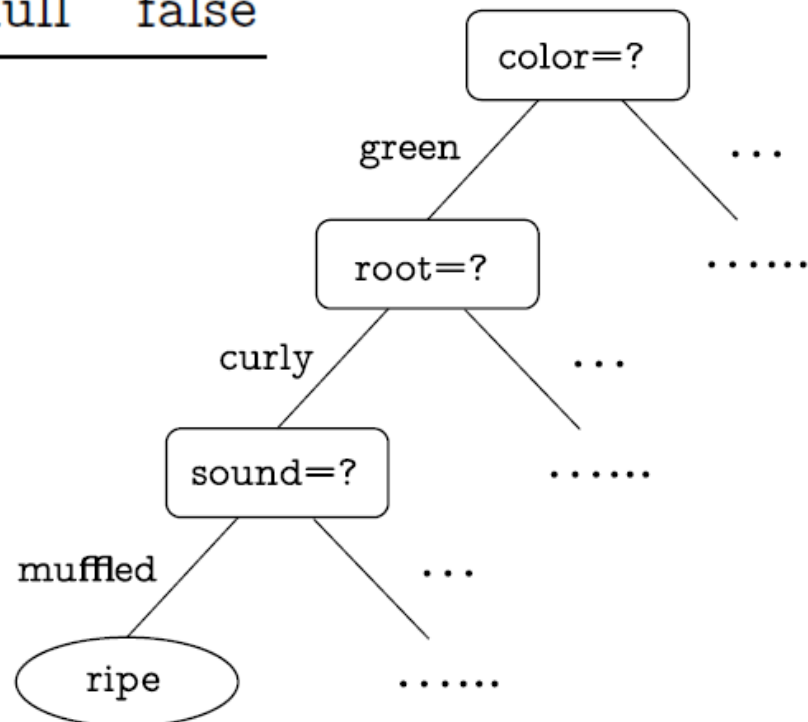
Algorithms used for ML

Decision Trees

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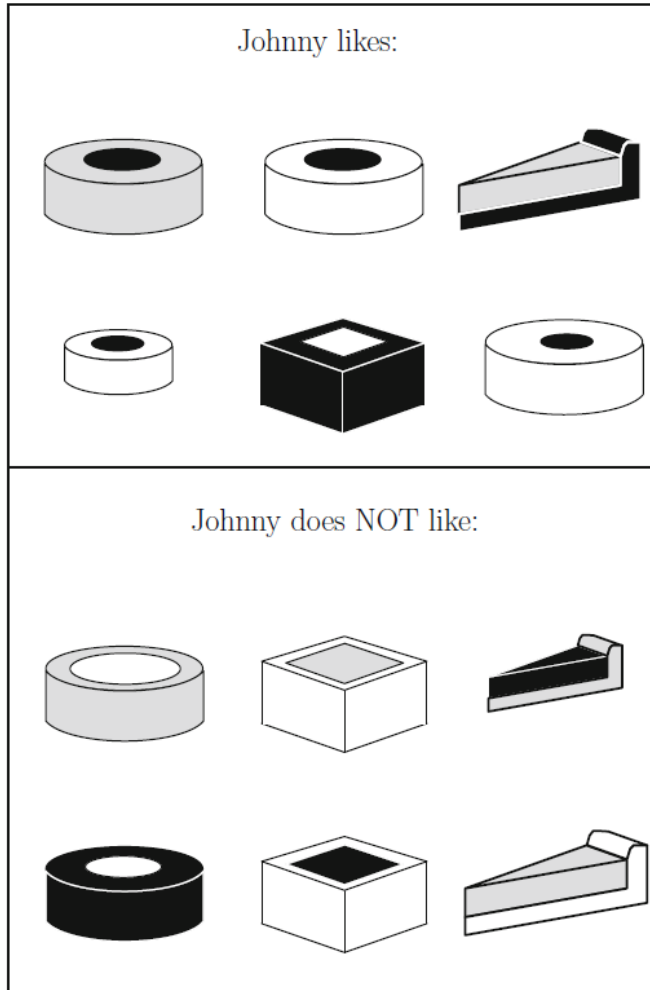
A toy watermelon data set

ID	color	root	sound	ripe
1	green	curly	muffled	true
2	dark	curly	muffled	true
3	green	straight	crisp	false
4	dark	slightly curly	dull	false



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Example Bayesian classifiers



Example	Shape	Crust		Filling		Class	# differences
		Size	Shade	Size	Shade		
x	Square	Thick	Gray	Thin	White	?	–
ex ₁	Circle	Thick	Gray	Thick	Dark	pos	3
ex ₂	Circle	Thick	White	Thick	Dark	pos	4
ex ₃	Triangle	Thick	Dark	Thick	Gray	pos	4
ex ₄	Circle	Thin	White	Thin	Dark	pos	4
ex ₅	Square	Thick	Dark	Thin	White	pos	1
ex ₆	Circle	Thick	White	Thin	Dark	pos	3
ex ₇	Circle	Thick	Gray	Thick	White	neg	2
ex ₈	Square	Thick	White	Thick	Gray	neg	3
ex ₉	Triangle	Thin	Gray	Thin	Dark	neg	3
ex ₁₀	Circle	Thick	Dark	Thick	White	neg	3
ex ₁₁	Square	Thick	White	Thick	Dark	neg	3
ex ₁₂	Triangle	Thick	White	Thick	Gray	neg	4

Example Bayesian classifiers

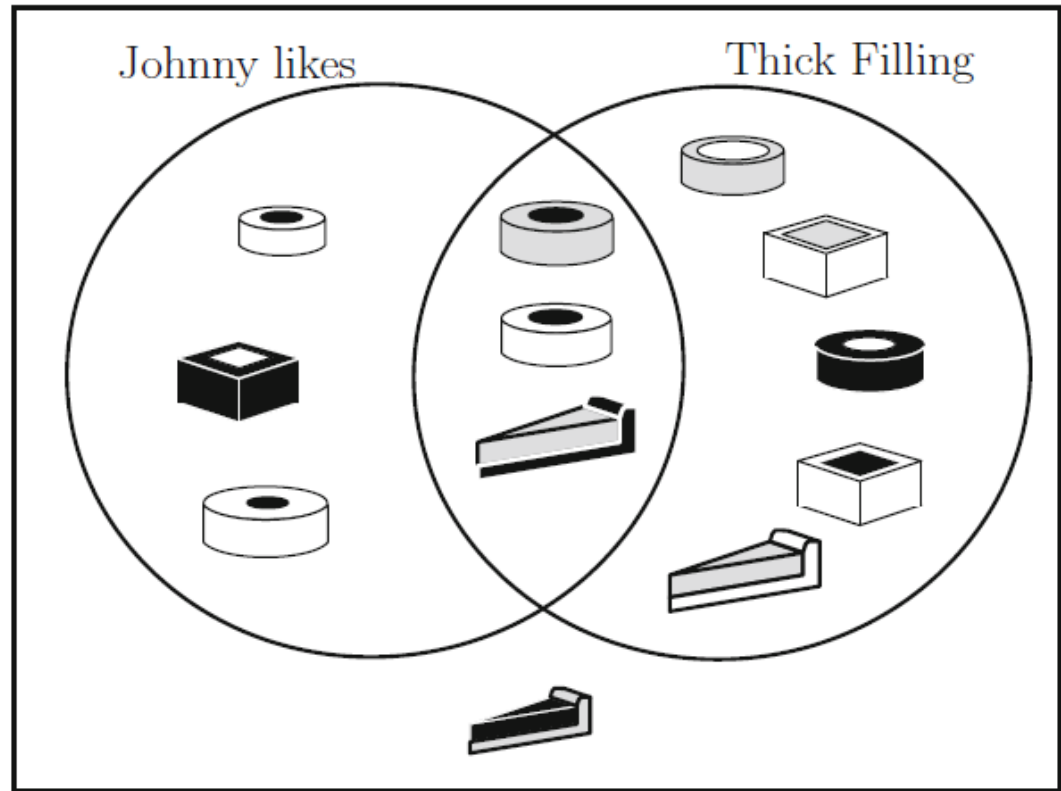
15

$$P(\text{pos}) = \frac{6}{12}$$

$$P(\text{thick}) = \frac{8}{12}$$

$$P(\text{pos}|\text{thick}) = \frac{3}{8}$$

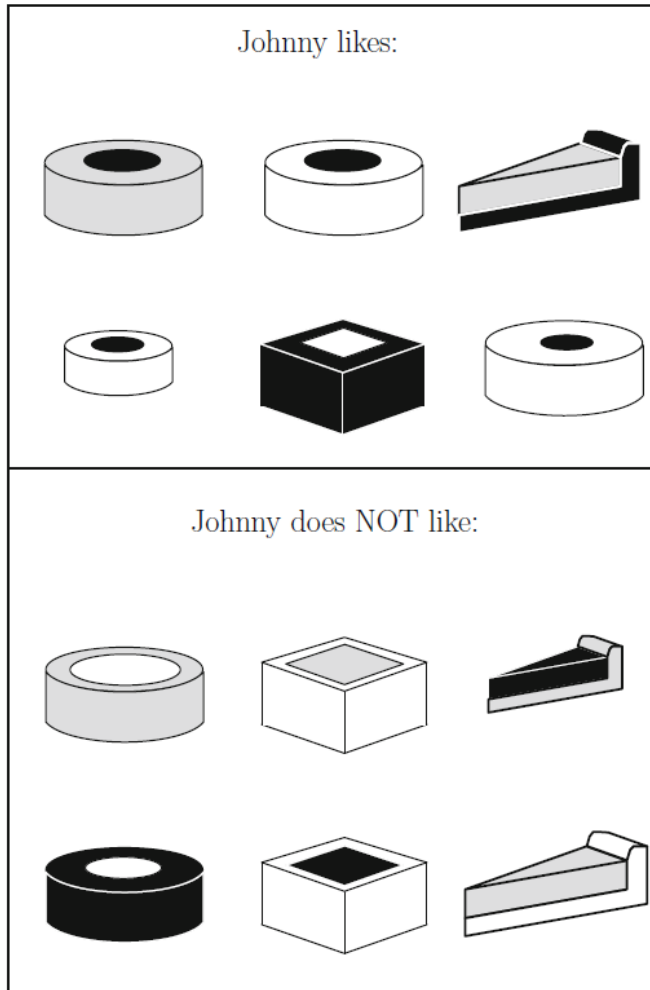
$$P(\text{thick}|\text{pos}) = \frac{3}{6}$$



$$P(\text{likes, thick}) = \frac{3}{12}$$

Example Nearest-Neighbor classifiers

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Example	Shape	Crust		Filling		Class	# differences
		Size	Shade	Size	Shade		
x	Square	Thick	Gray	Thin	White	?	–
ex ₁	Circle	Thick	Gray	Thick	Dark	pos	3
ex ₂	Circle	Thick	White	Thick	Dark	pos	4
ex ₃	Triangle	Thick	Dark	Thick	Gray	pos	4
ex ₄	Circle	Thin	White	Thin	Dark	pos	4
ex ₅	Square	Thick	Dark	Thin	White	pos	1
ex ₆	Circle	Thick	White	Thin	Dark	pos	3
ex ₇	Circle	Thick	Gray	Thick	White	neg	2
ex ₈	Square	Thick	White	Thick	Gray	neg	3
ex ₉	Triangle	Thin	Gray	Thin	Dark	neg	3
ex ₁₀	Circle	Thick	Dark	Thick	White	neg	3
ex ₁₁	Square	Thick	White	Thick	Dark	neg	3
ex ₁₂	Triangle	Thick	White	Thick	Gray	neg	4

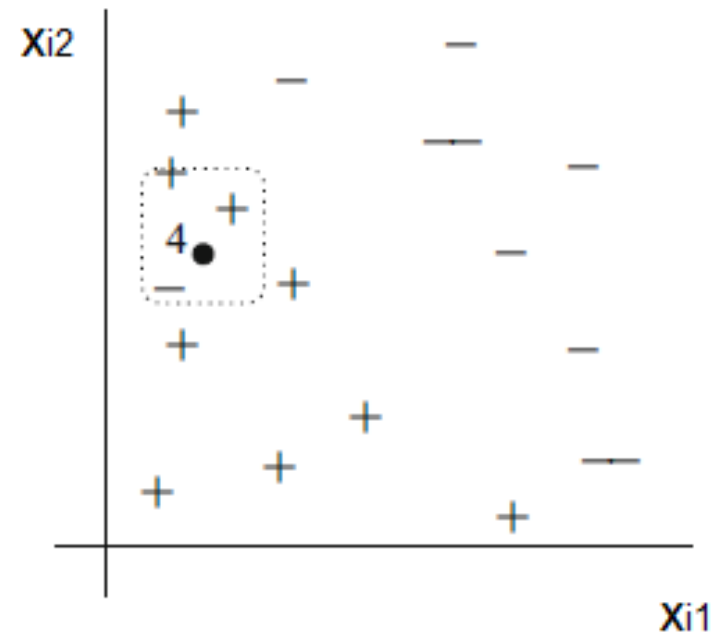
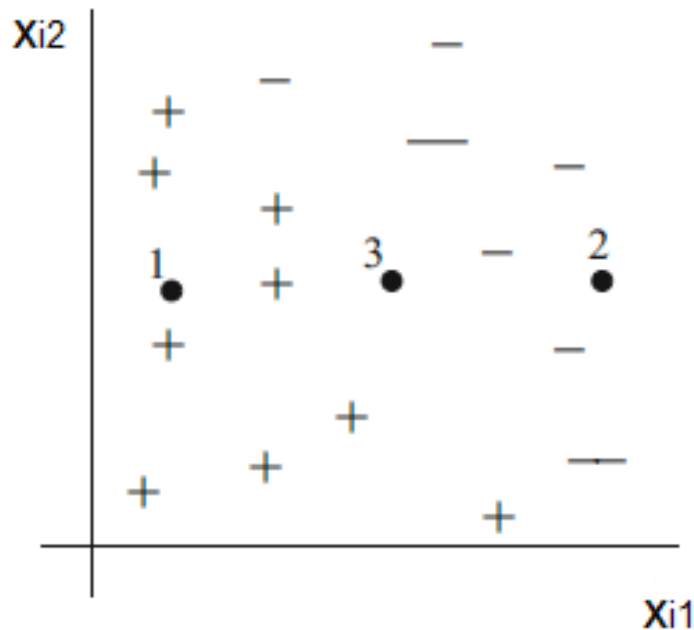
→ new pie x is most similar to ex₅
 → positive → johnny likes it

Exercise Nearest-Neighbor classifiers

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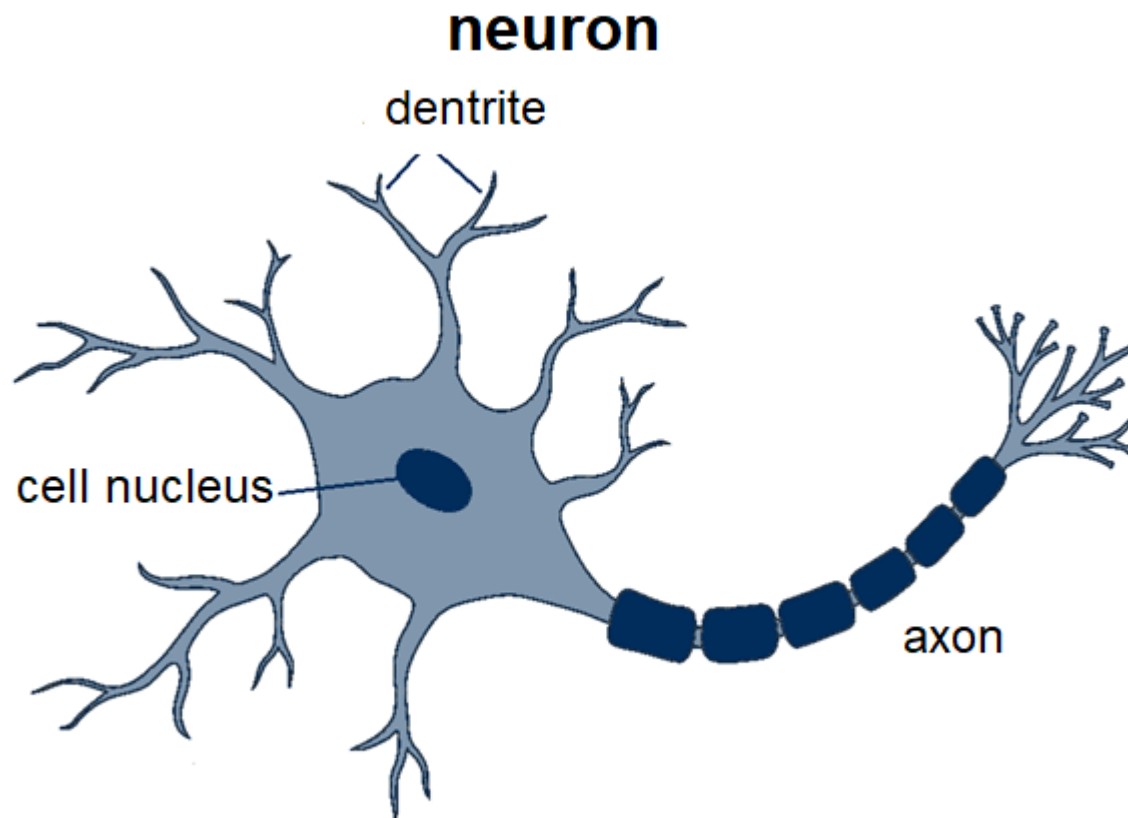
+/- are two classes

- is unknown sample



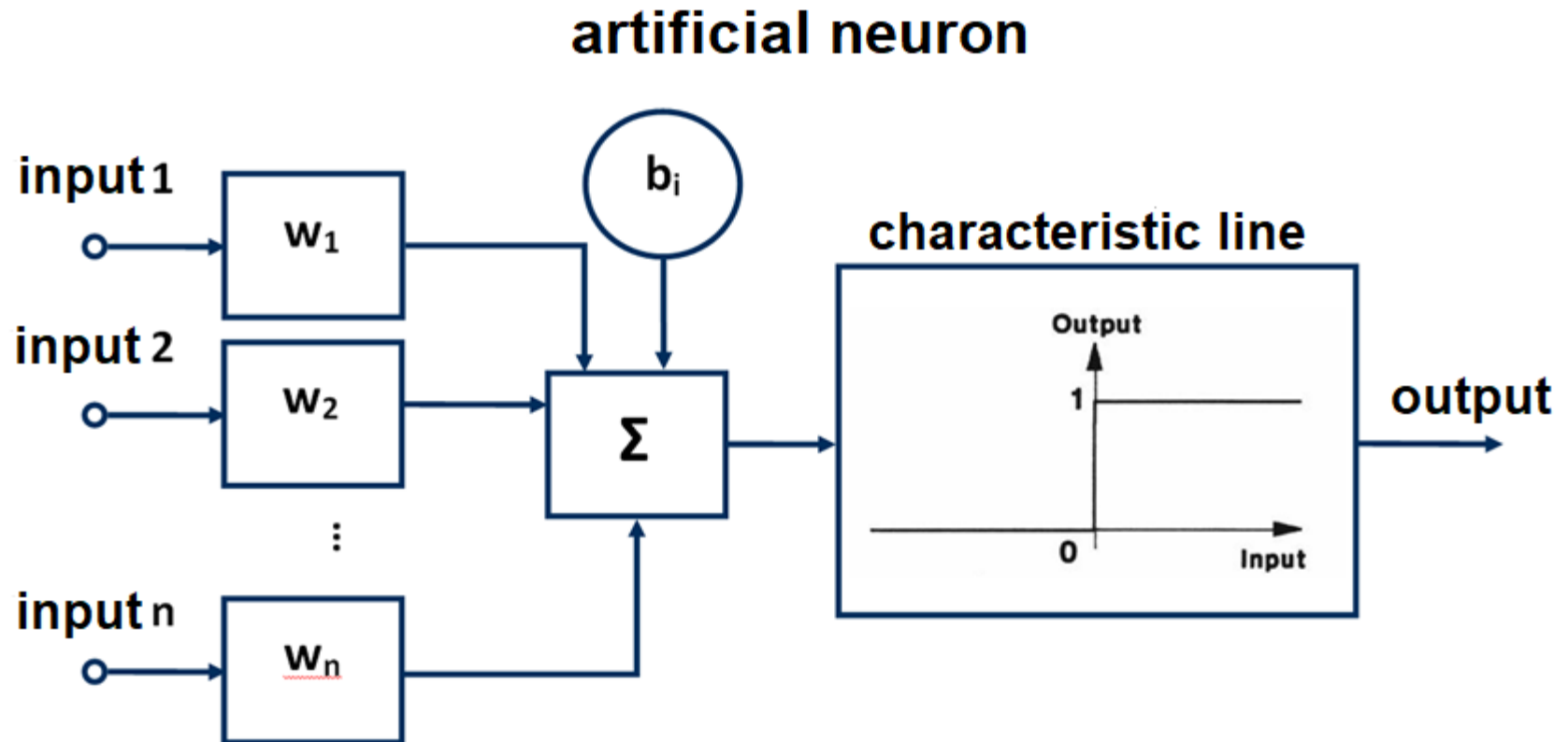
Neural Networks – natural neuron

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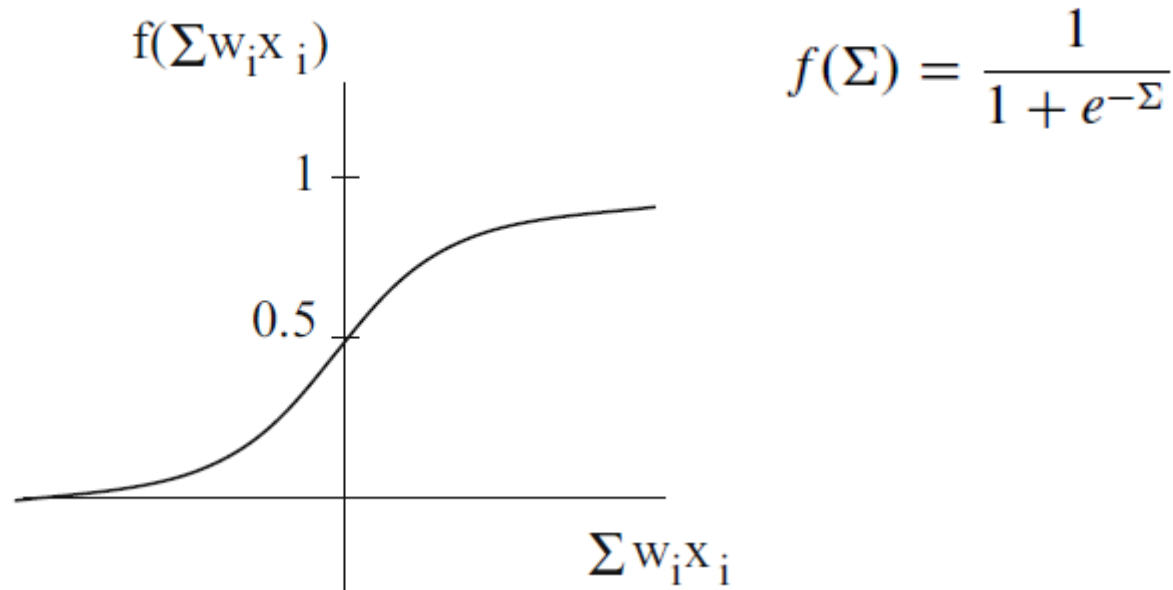


Neural Networks – artificial neuron

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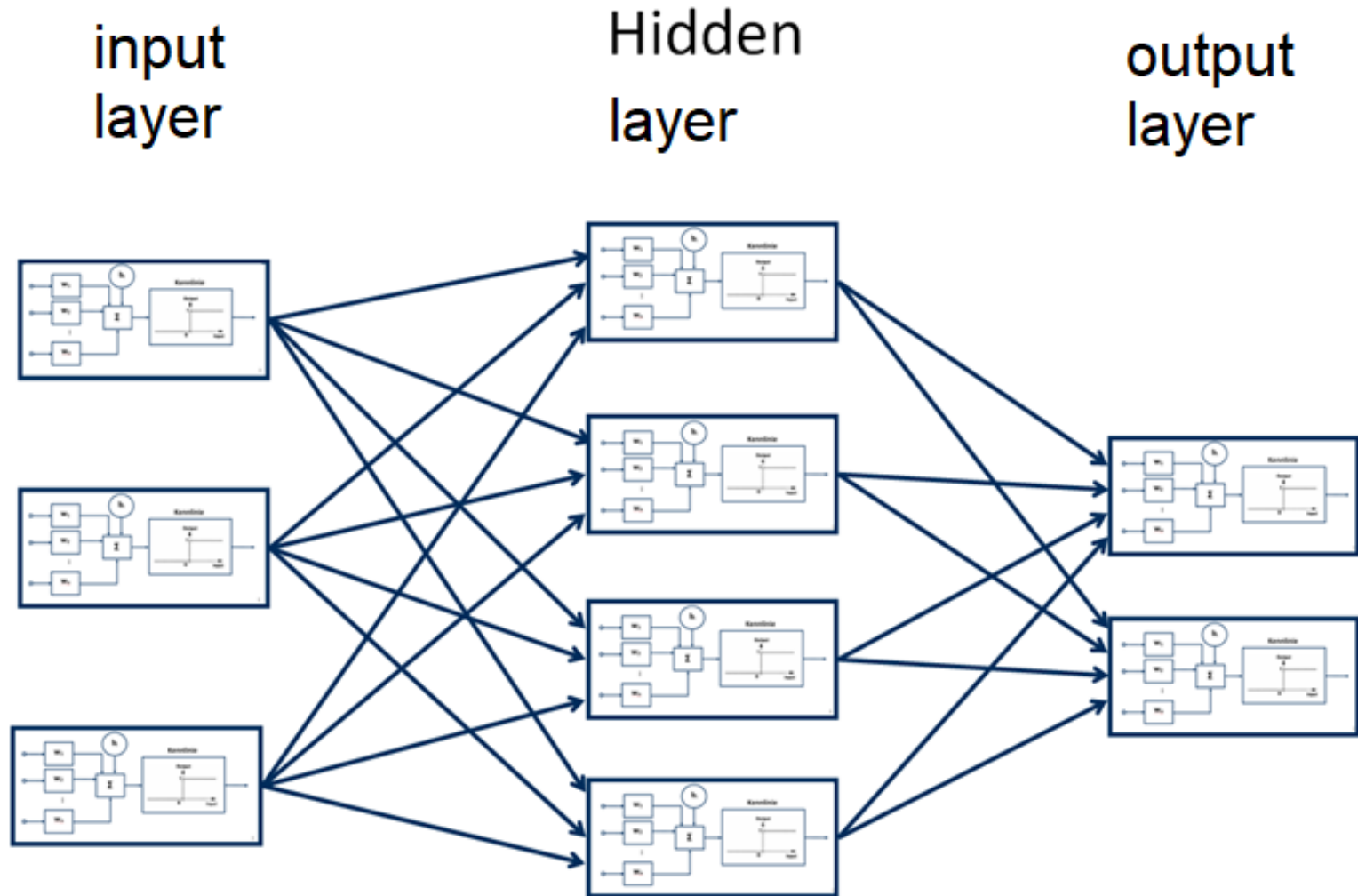
Popular transfer function sigmoid



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Neural Networks - layers

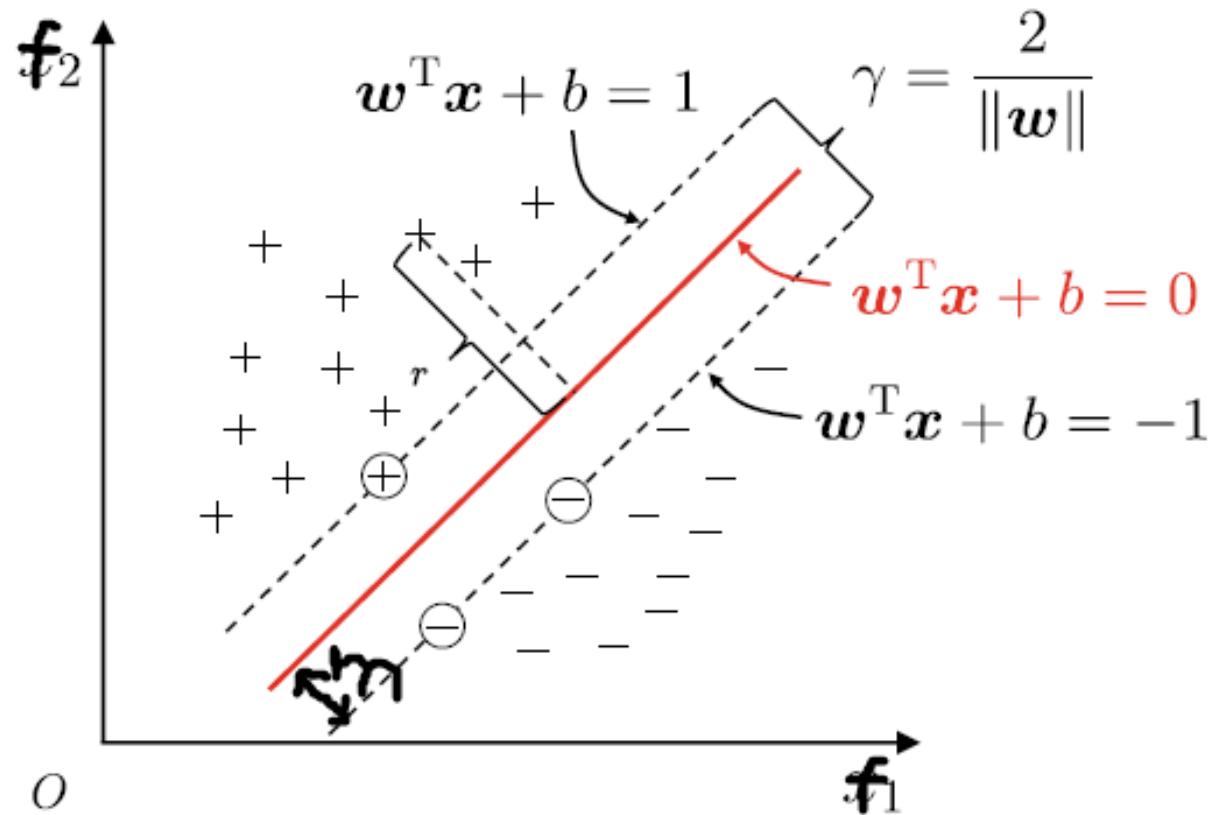
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$$y_i = f \left(\sum_j w_{ji}^{(1)} f \left(\sum_l w_{lj}^{(2)} x_l \right) \right)$$

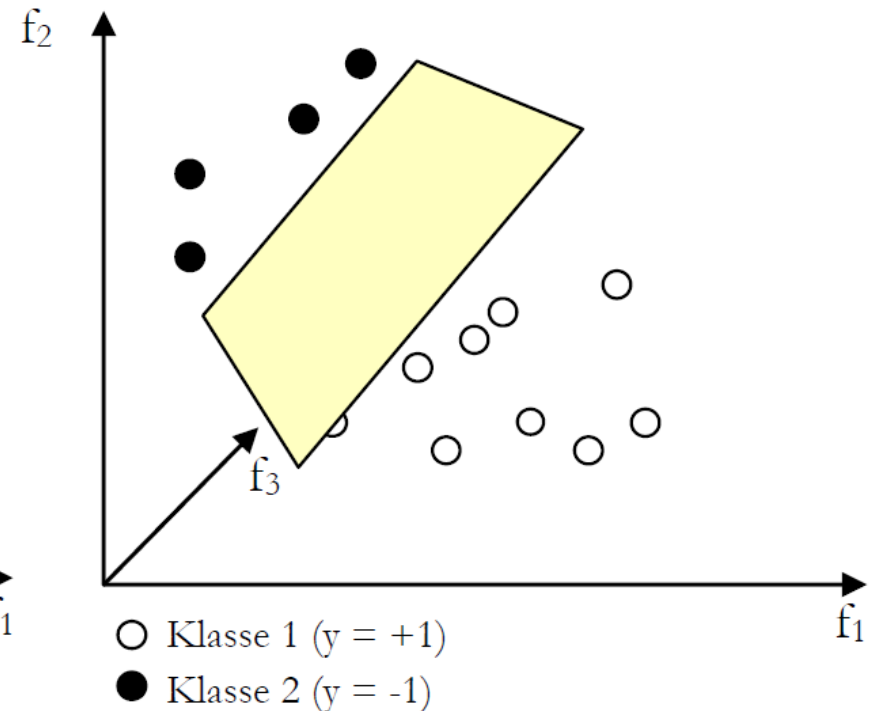
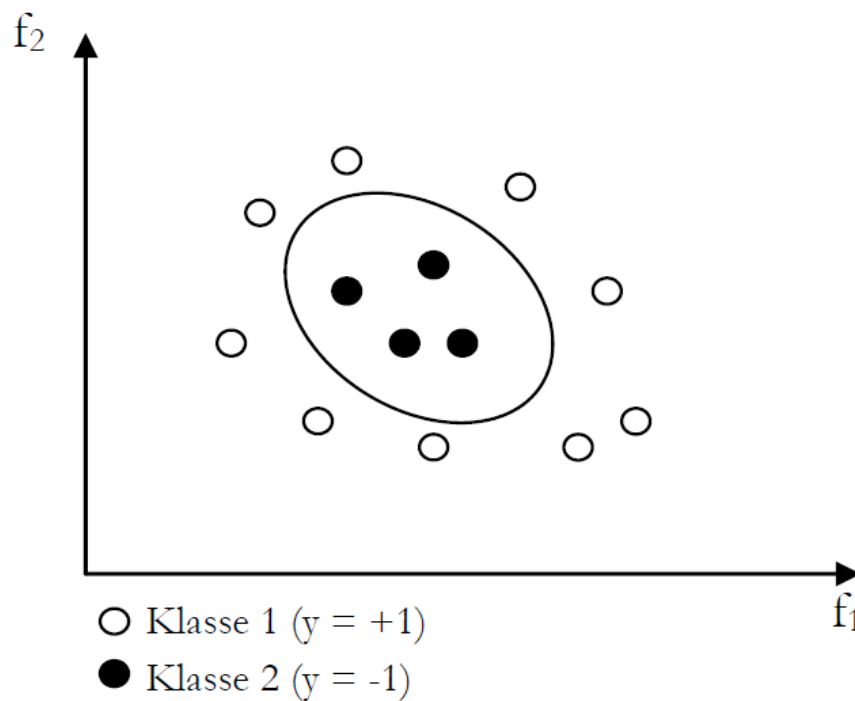
Support Vector Machines

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Increasing dimensions for linear separating:



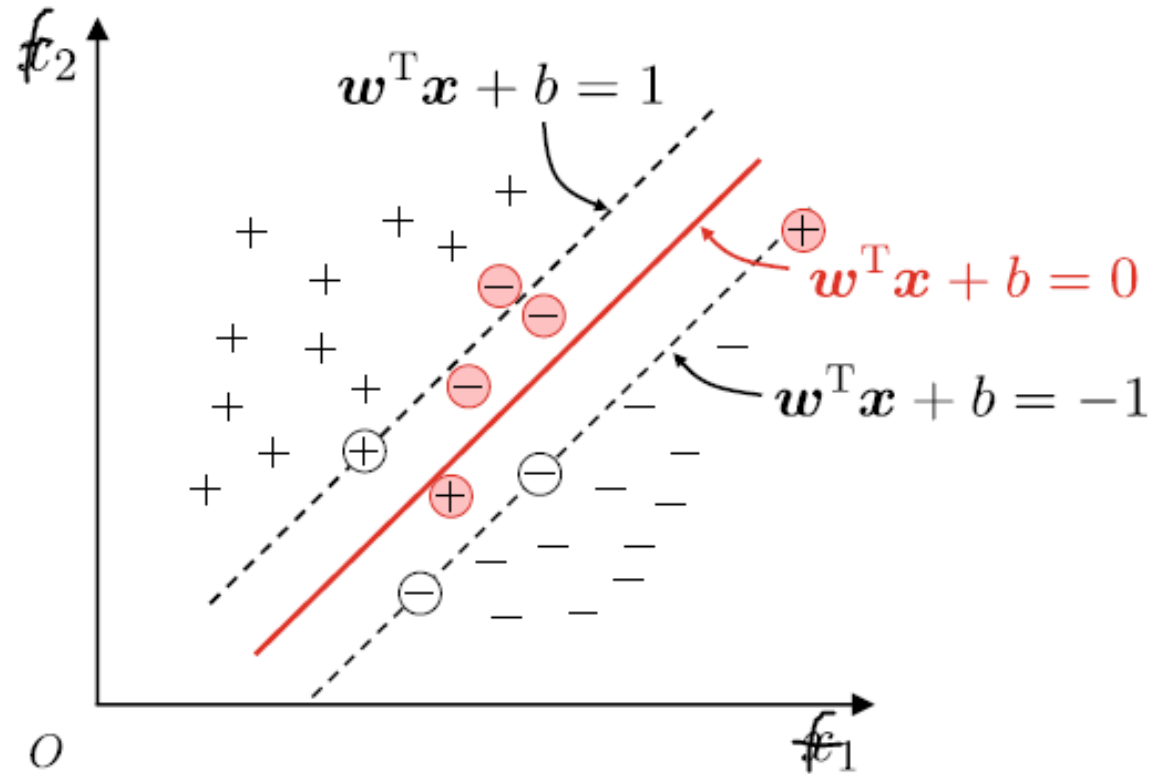
Kernel functions

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Name	Expression	Parameters
Linear kernel	$\kappa(\mathbf{x}_i, \mathbf{x}_j) = \mathbf{x}_i^\top \mathbf{x}_j$	
Polynomial kernel	$\kappa(\mathbf{x}_i, \mathbf{x}_j) = (\mathbf{x}_i^\top \mathbf{x}_j)^d$	$d \geq 1$ is the degree of the polynomial
Gaussian kernel	$\kappa(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\frac{\ \mathbf{x}_i - \mathbf{x}_j\ ^2}{2\sigma^2}\right)$	$\sigma > 0$ is the width of the Gaussian kernel
Laplacian kernel	$\kappa(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\frac{\ \mathbf{x}_i - \mathbf{x}_j\ }{\sigma}\right)$	$\sigma > 0$
Sigmoid kernel	$\kappa(\mathbf{x}_i, \mathbf{x}_j) = \tanh(\beta \mathbf{x}_i^\top \mathbf{x}_j + \theta)$	$\tanh$ is the hyperbolic tangent function, $\beta > 0$, $\theta < 0$

Soft Margin

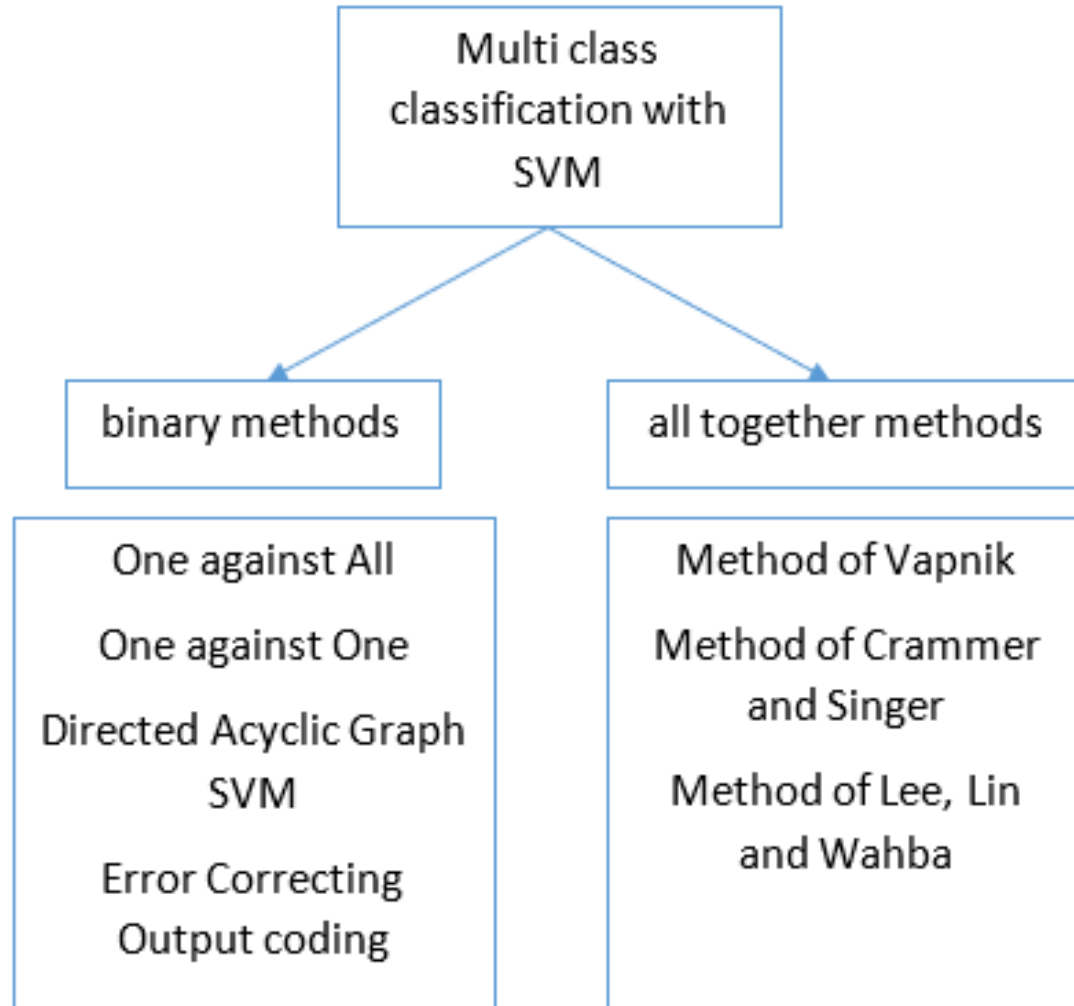
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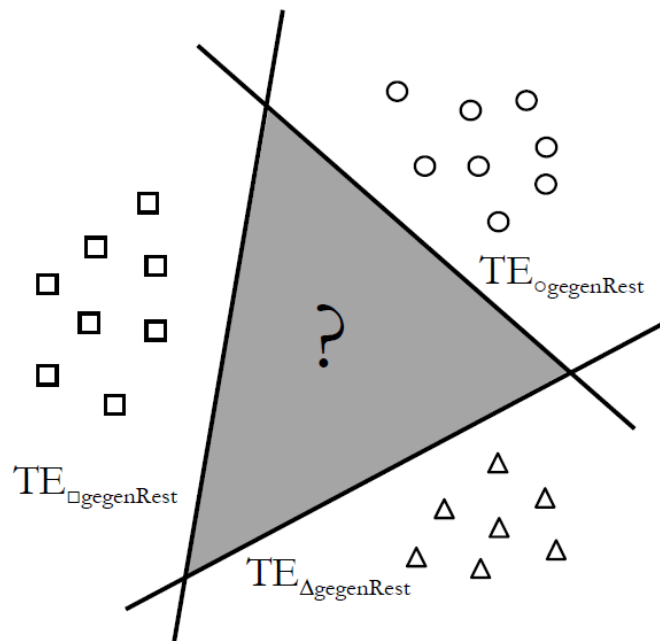
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Multi class SVM

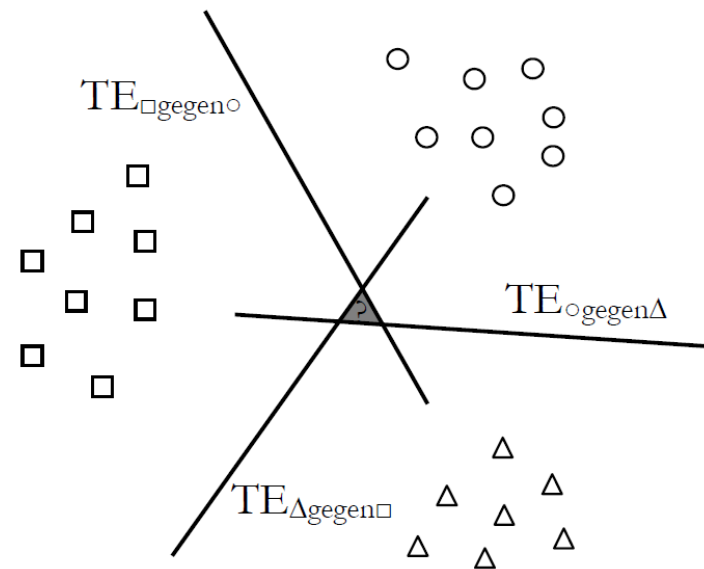
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One against All

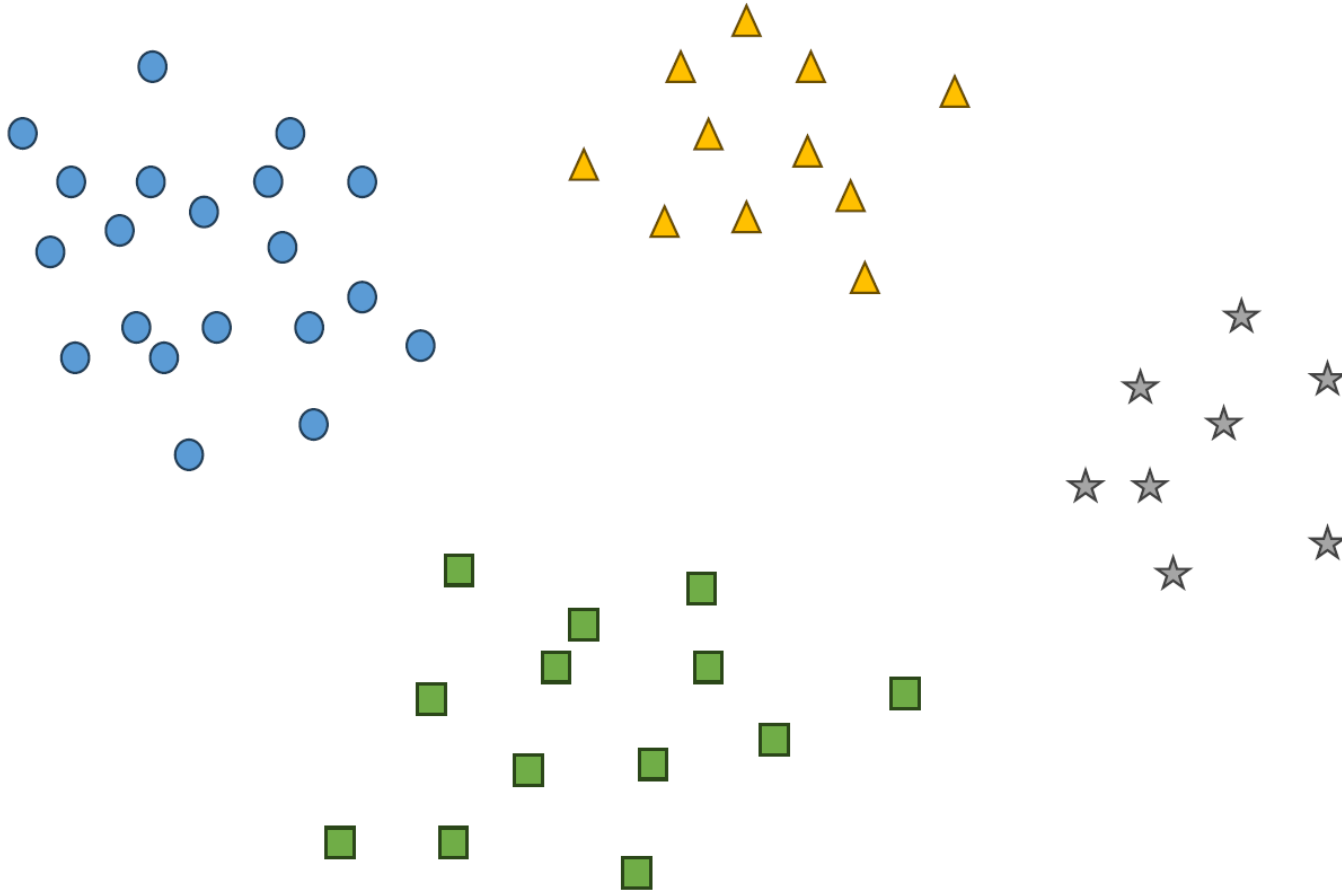


One against One



Exercise Multiclass SVM OAA

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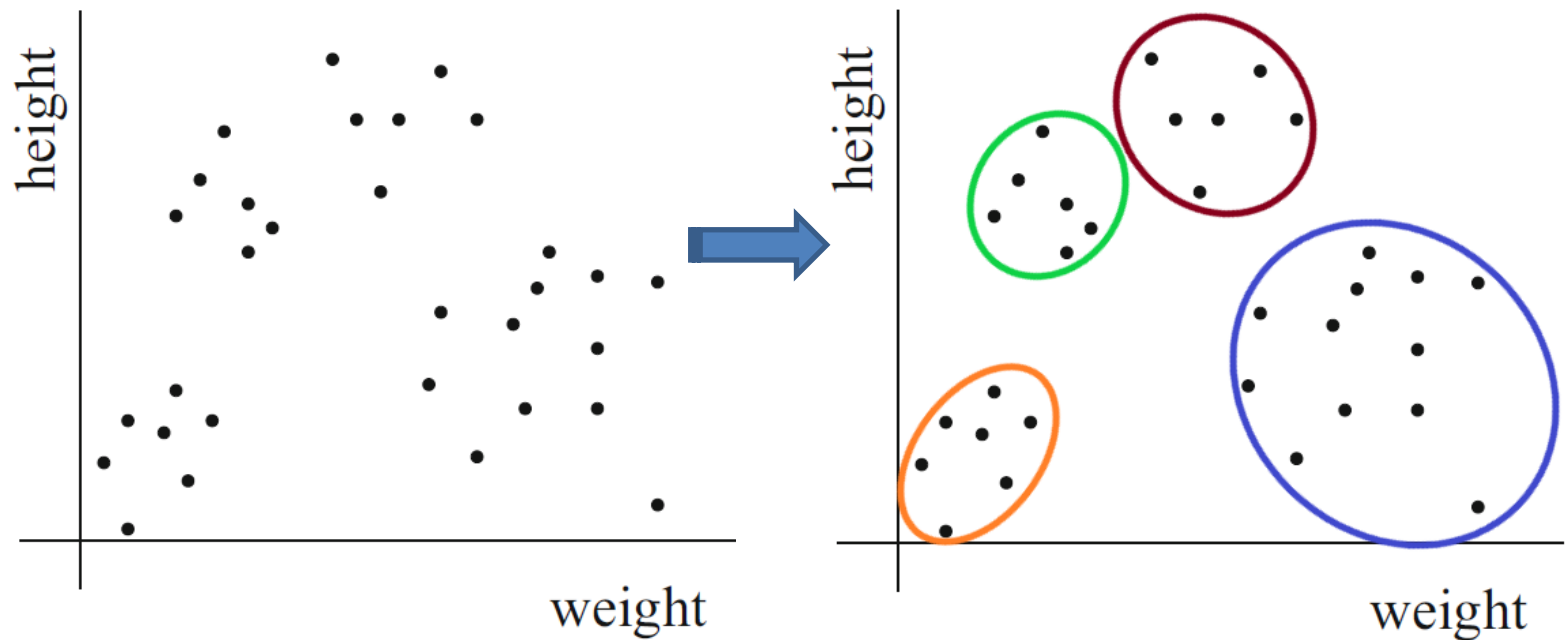


Methods of ML

Cluster Analysis

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Simple example: dataset of samples with 2 feature 'weight' and 'height' → grouped into 4 cluster

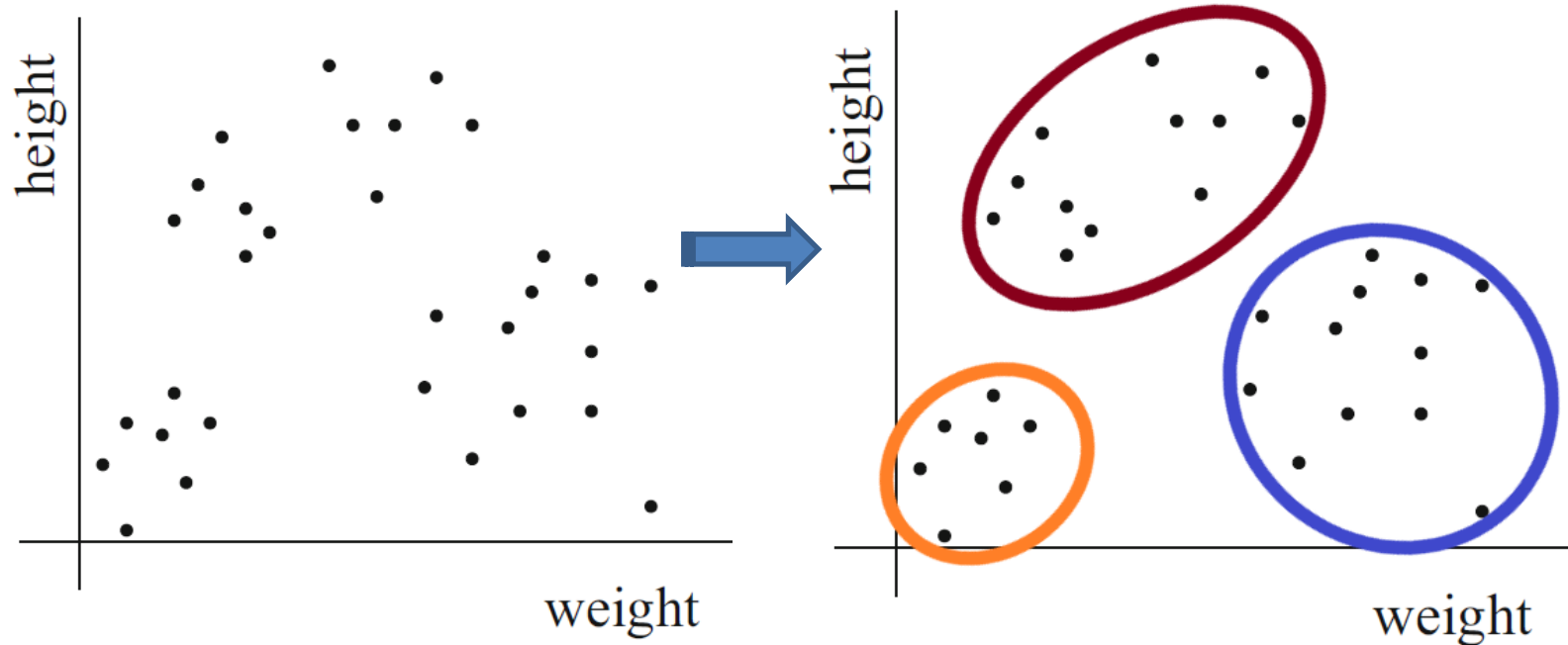


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Cluster Analysis

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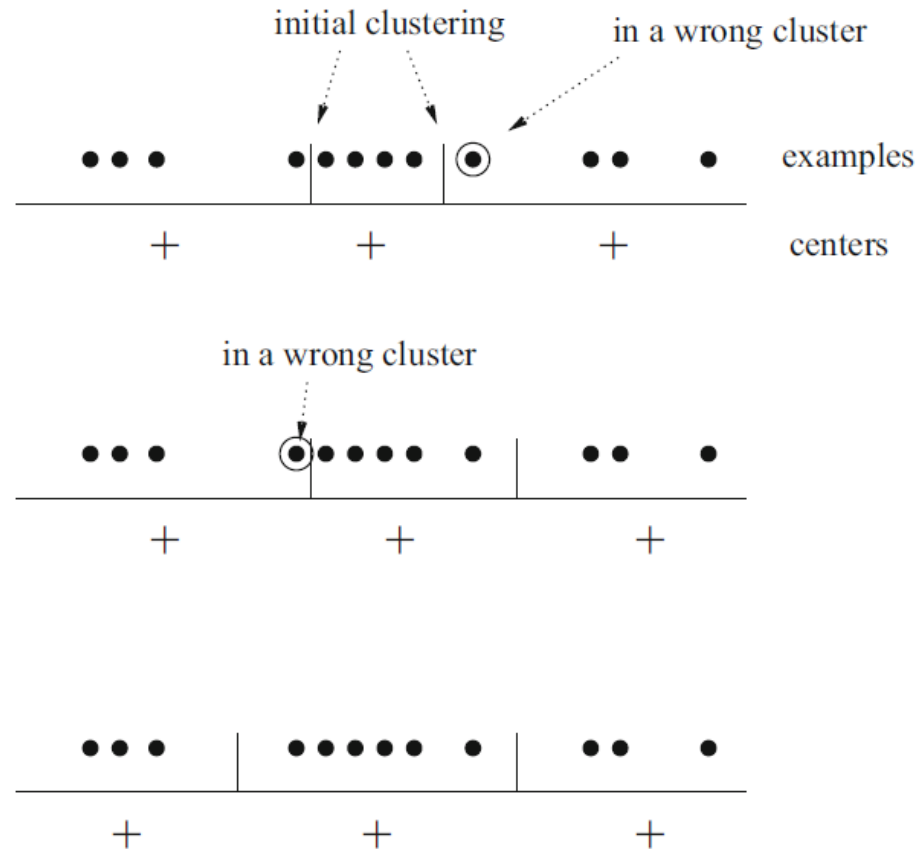
Simple example: dataset of samples with 2 feature 'weight' and 'height' → grouped into 3 cluster



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Cluster Analysis with kmeans

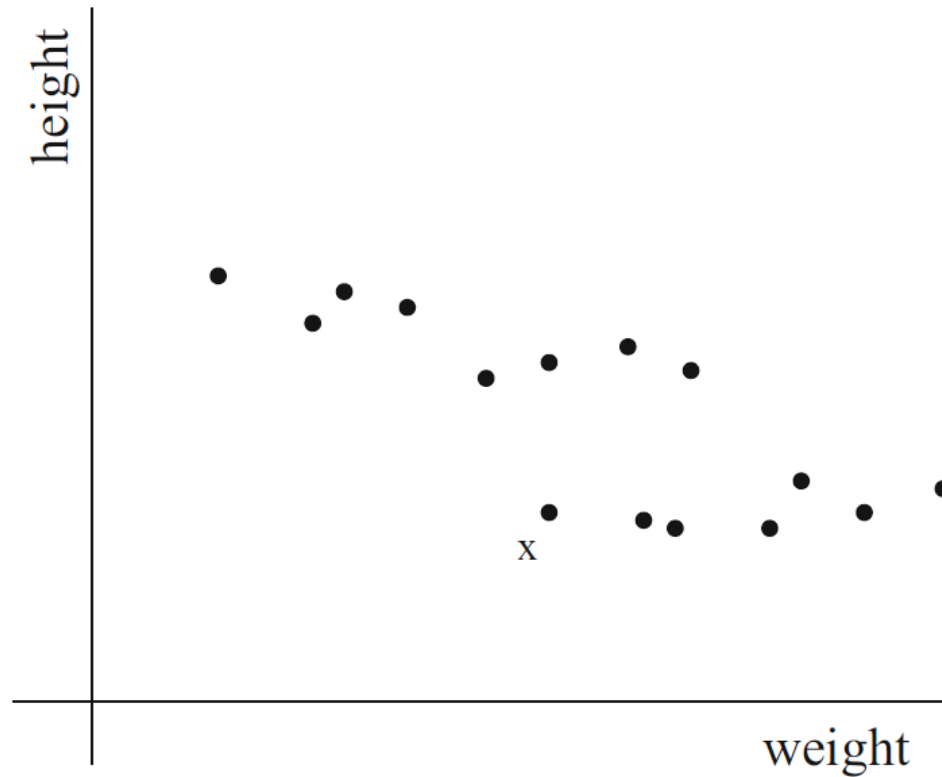
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Example non-convex cluster

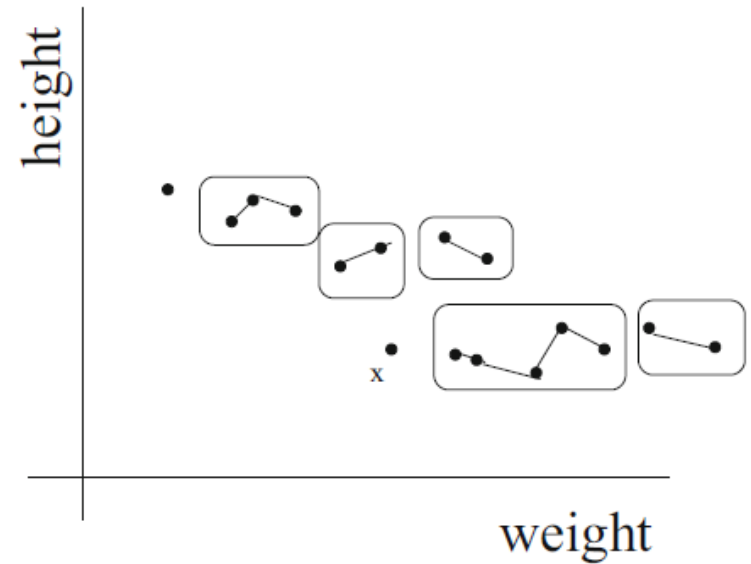
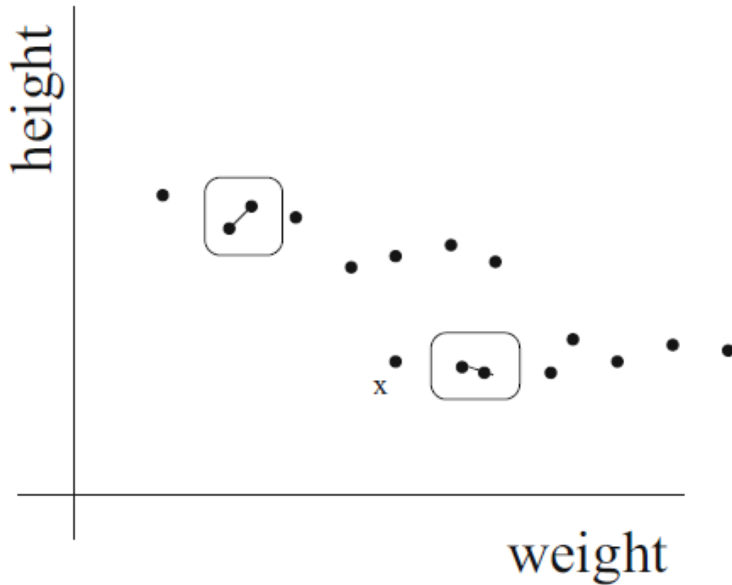
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Example non-convex cluster

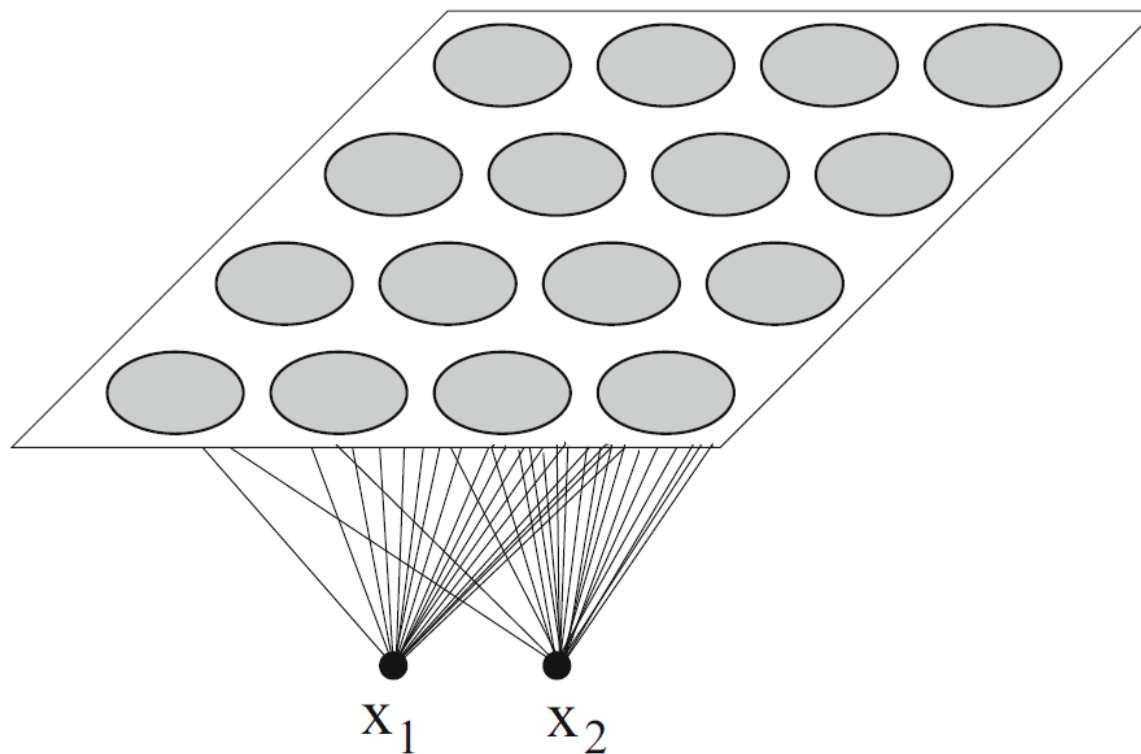
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Self-Organizing Feature Maps

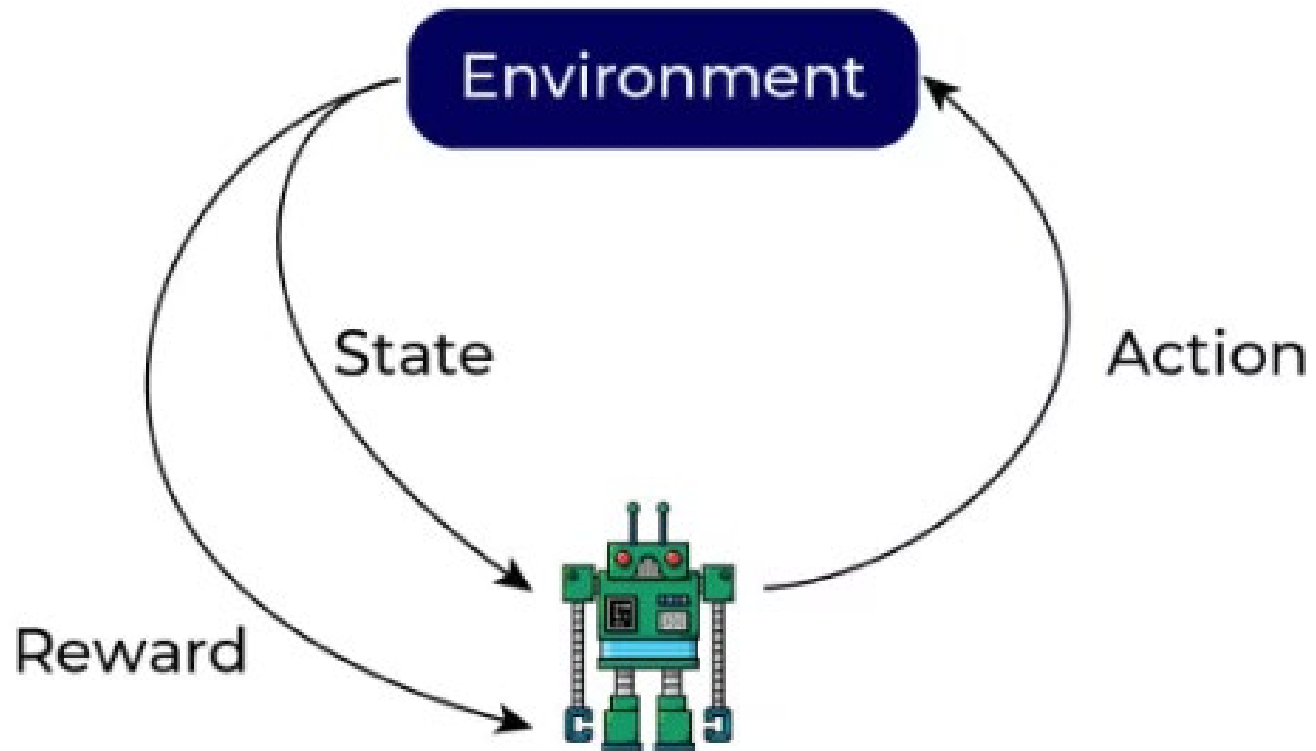
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Reinforcement learning

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<https://datasolut.com/reinforcement-learning/>

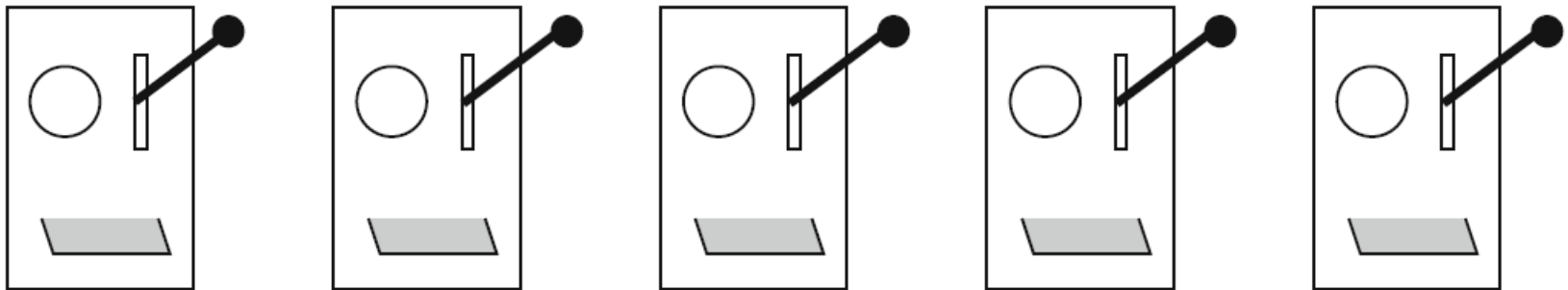
Example N-armed bandit

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5 slot machines →

Which is the most promising machine to increase our gain? →

find out average returns of each machine



Example RL

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Game Board:



Current state (s):
 $\begin{matrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{matrix}$

Q Table:

$\gamma = 0.95$

	$\begin{matrix} 0 & 0 & 0 \\ 1 & 0 & 0 \end{matrix}$	$\begin{matrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{matrix}$	$\begin{matrix} 0 & 0 & 0 \\ 0 & 0 & 1 \end{matrix}$	$\begin{matrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{matrix}$	$\begin{matrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{matrix}$	$\begin{matrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{matrix}$
↑	0.2	0.3	1.0	-0.22	-0.3	0.0
↓	-0.5	-0.4	-0.2	-0.04	-0.02	0.0
→	0.21	0.4	-0.3	0.5	1.0	0.0
←	-0.6	-0.1	-0.1	-0.31	-0.01	0.0

<https://datasolut.com/reinforcement-learning/>

Example RL

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→ State:

A	B	C	D	E	F
0 0 0	0 0 0	0 0 0	1 0 0	0 1 0	0 0 1
1 0 0	0 1 0	0 0 1	0 0 0	0 0 0	0 0 0

→ Actions replaced by final state which will be reached by motion [up,down,right, left]

Example RL

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$Q_0 =$	A	B	C	D	E	F
A	0	0	0	0	0	0
B	0	0	0	0	0	0
C	0	0	0	0	0	0
D	0	0	0	0	0	0
E	0	0	0	0	0	0
F	0	0	0	0	0	0

R =	A	B	C	D	E	F
A	-	0	-	0	-	-
B	0	-	0	-	0	-
C	-	0	-	-	-	100
D	0	-	-	-	0	-
E	-	0	-	0	-	100
F	-	-	0	-	0	100

- Q-table at beginning initialized to 0
- - ... no move to this state possible or stay at this state make no sense
- 0 ... no reward
- 100 ... reward to reach the goal
- $\alpha=1$
- $\gamma=0.95$

Example RL

- Q1 after first iteration (path $A \rightarrow B \rightarrow C \rightarrow F$)

$Q_1 =$	A	B	C	D	E	F
A	0	0	0	0	0	0
B	0	0	0	0	0	0
C	0	0	0	0	0	100
D	0	0	0	0	0	0
E	0	0	0	0	0	0
F	0	0	0	0	0	0

R =	A	B	C	D	E	F
A	-	0	-	0	-	-
B	0	-	0	-	0	-
C	-	0	-	-	-	100
D	0	-	-	-	0	-
E	-	0	-	0	-	100
F	-	-	0	-	0	100

Example RL

- Q2 after second iteration (path $D \rightarrow E \rightarrow F$)

Q ₂ =	A	B	C	D	E	F
A	0	0	0	0	0	0
B	0	0	0	0	0	0
C	0	0	0	0	0	100
D	0	0	0	0	0	0
E	0	0	0	0	0	100
F	0	0	0	0	0	0

R=	A	B	C	D	E	F
A	-	0	-	0	-	-
B	0	-	0	-	0	-
C	-	0	-	-	-	100
D	0	-	-	-	0	-
E	-	0	-	0	-	100
F	-	-	0	-	0	100

Example RL

- Q3 after first part of third iteration (path $B \rightarrow E \rightarrow D \rightarrow A$)

$Q_3 =$	A	B	C	D	E	F
A	0	0	0	0	0	0
B	0	0	0	0	95	0
C	0	0	0	0	0	100
D	0	0	0	0	0	0
E	0	0	0	0	0	100
F	0	0	0	0	0	0

R =	A	B	C	D	E	F
A	-	0	-	0	-	-
B	0	-	0	-	0	-
C	-	0	-	-	-	100
D	0	-	-	-	0	-
E	-	0	-	0	-	100
F	-	-	0	-	0	100

- Go on with $B \rightarrow C \rightarrow F$ and determine Q-table!

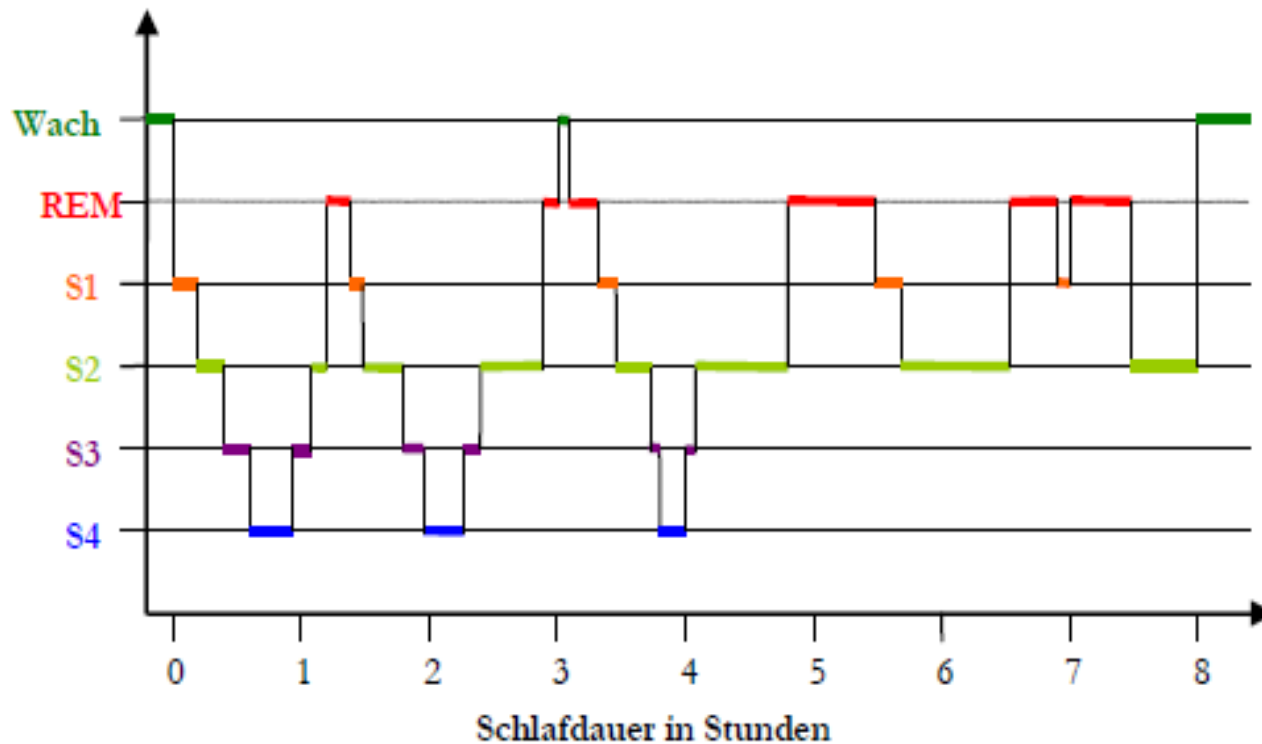
Data Science and ML

- **Missing data**
- **Invalid data**
- **Unbalance data**
- **Noise**

Example unbalanced data

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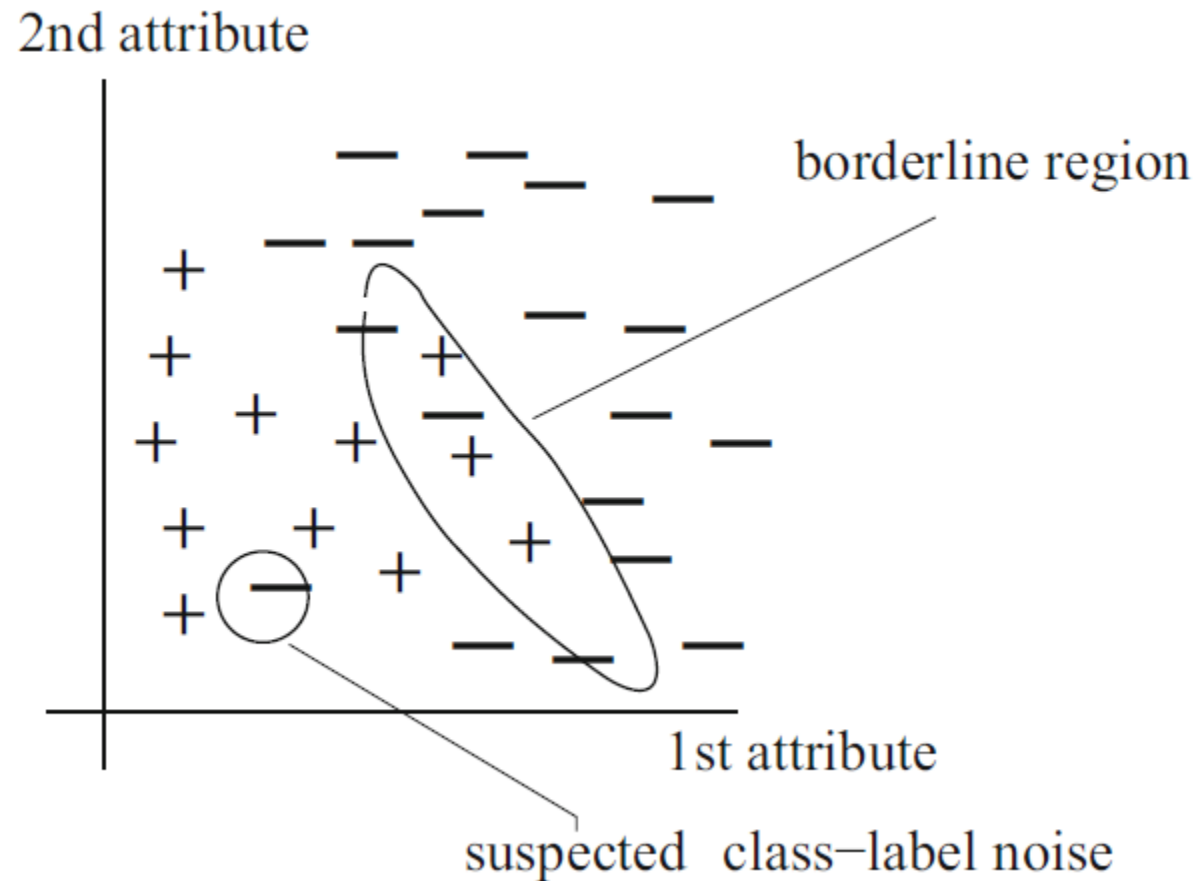
Sleep classification with EEG:



Normal sleep profile (sleep stages over night)

Example noisy data

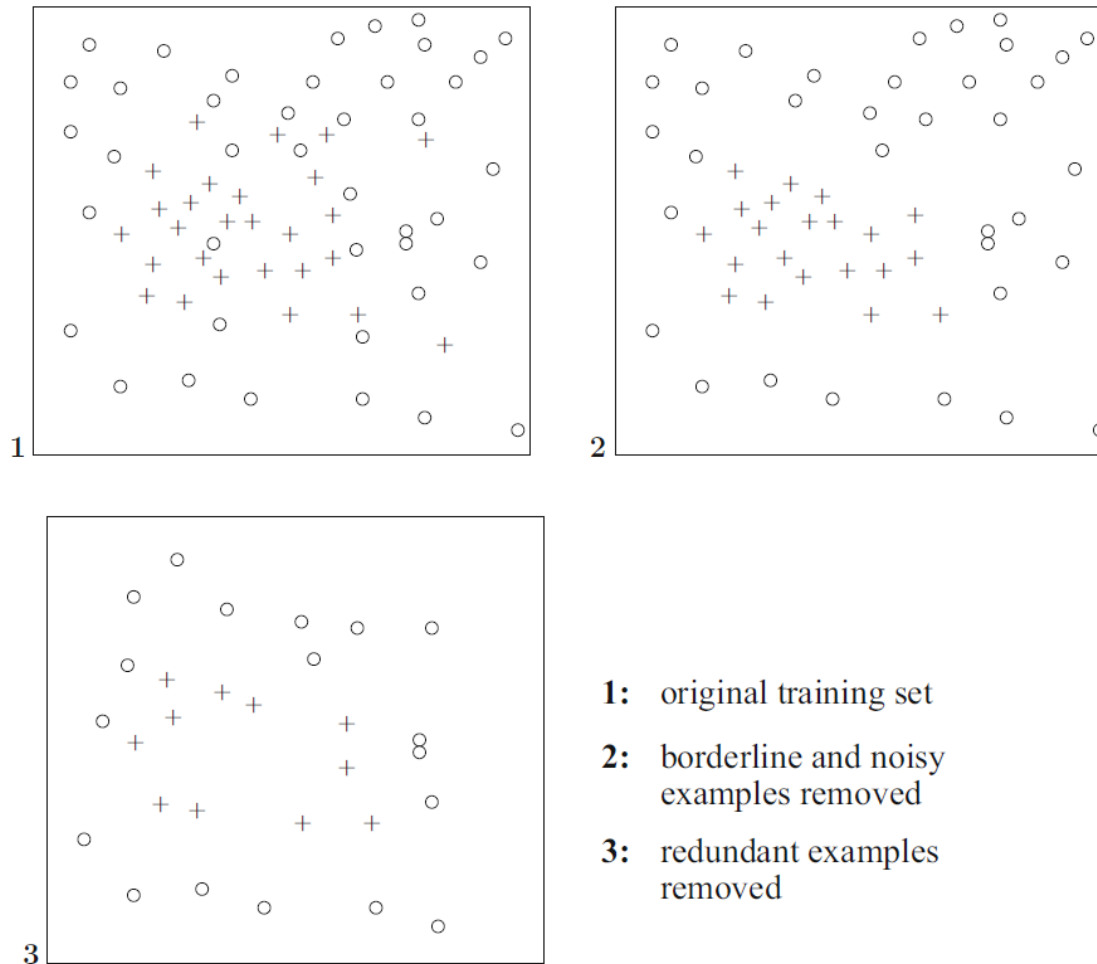
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Kubat, An Introduction to Machine Learning, 2017

Example pre-processing of data

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Kubat, An Introduction to Machine Learning, 2017

Quality values of ML Algorithm

Error rate E

$$E = \frac{\sum wrong}{N}$$

Accuracy

$$A = 1 - e = \frac{\sum right}{N}$$

Confusion matrix

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- 2class problems

		Labels returned by classifier	
		pos	neg
Real labels	Pos	N_{TP}	N_{FN}
	Neg	N_{FP}	N_{TN}

$$E = \frac{N_{FP} + N_{FN}}{N_{TP} + N_{FP} + N_{FN} + N_{TN}}$$

$$A = \frac{N_{TP} + N_{TN}}{N_{TP} + N_{FP} + N_{FN} + N_{TN}}$$

N_{TP} – number of samples true positive

N_{FP} – number of samples false positive

N_{FN} – number of samples false negative

N_{TN} – number of samples true negative

Quality values

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Error rate E

$$E = \frac{\sum wrong}{N}$$

Accuracy

$$A = 1 - e = \frac{\sum right}{N}$$

Precision

$$P = \frac{N_{TP}}{N_{TP} + N_{FP}}$$

Recall

$$R = \frac{N_{TP}}{N_{TP} + N_{FN}} = \frac{N_{TP}}{N_P}$$

Specificity

$$S = \frac{N_{TN}}{N_{TN} + N_{FP}} = \frac{N_{TN}}{N_N}$$

Balanced accuracy

$$A_b = \frac{\frac{N_{TP}}{N_P} + \frac{N_{TN}}{N_N}}{2}$$

Balanced precision

$$r = \frac{N_P}{N_N}$$

$$P_b = \frac{\frac{N_{TP}}{N_P}}{\frac{N_{TP}}{N_P} + \frac{1}{r} * \frac{N_{FP}}{N_N}}$$

Examples 2-class problems

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Perfect classifier:

		Labels returned by classifier	
		pos	neg
Real labels	Pos (1960)	1960	0
	Neg (40)	0	40

Useless classifier:

		Labels returned by classifier	
		pos	neg
Real labels	Pos (1960)	1960	0
	Neg (40)	40	0

Balanced dataset:

		Labels returned by classifier	
		pos	neg
Real labels	Pos (500)	365	135
	Neg (500)	88	412

Exercise 2-class problem

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		Labels returned by classifier	
		pos	neg
Real labels	Pos (150)	145	5
	Neg (50)	25	25

How many samples classified right? Calculate E and A for this classifier. Calculate Precision, Recall and Specificity. Is it a balanced dataset? If it is, calculate balanced accuracy and precision. Do the classifier show a good performance?

- M-class problems

		Labels returned by classifier				
		C1	C2	C3	...	Cm
Real labels	C1	N₁₁	N ₁₂	N ₁₃		N _{1m}
	C2	N ₂₁	N₂₂	N ₂₃		N _{2m}
	C3	N ₃₁	N ₃₂	N₃₃		N _{3m}
	...					
	Cm	N _{m1}	N _{m2}	N _{m3}		N_{mm}

$$A = \frac{\sum \text{right}}{N} = \frac{\sum_{i=1}^m N_{ii}}{\sum_{i=1}^m \sum_{j=1}^m N_{ij}}$$

$$ACA = \frac{1}{m} \sum \frac{\text{right}_m}{N_m} = \frac{1}{m} \sum_{i=1}^m \frac{N_{ii}}{\sum_{j=1}^m N_{ij}}$$

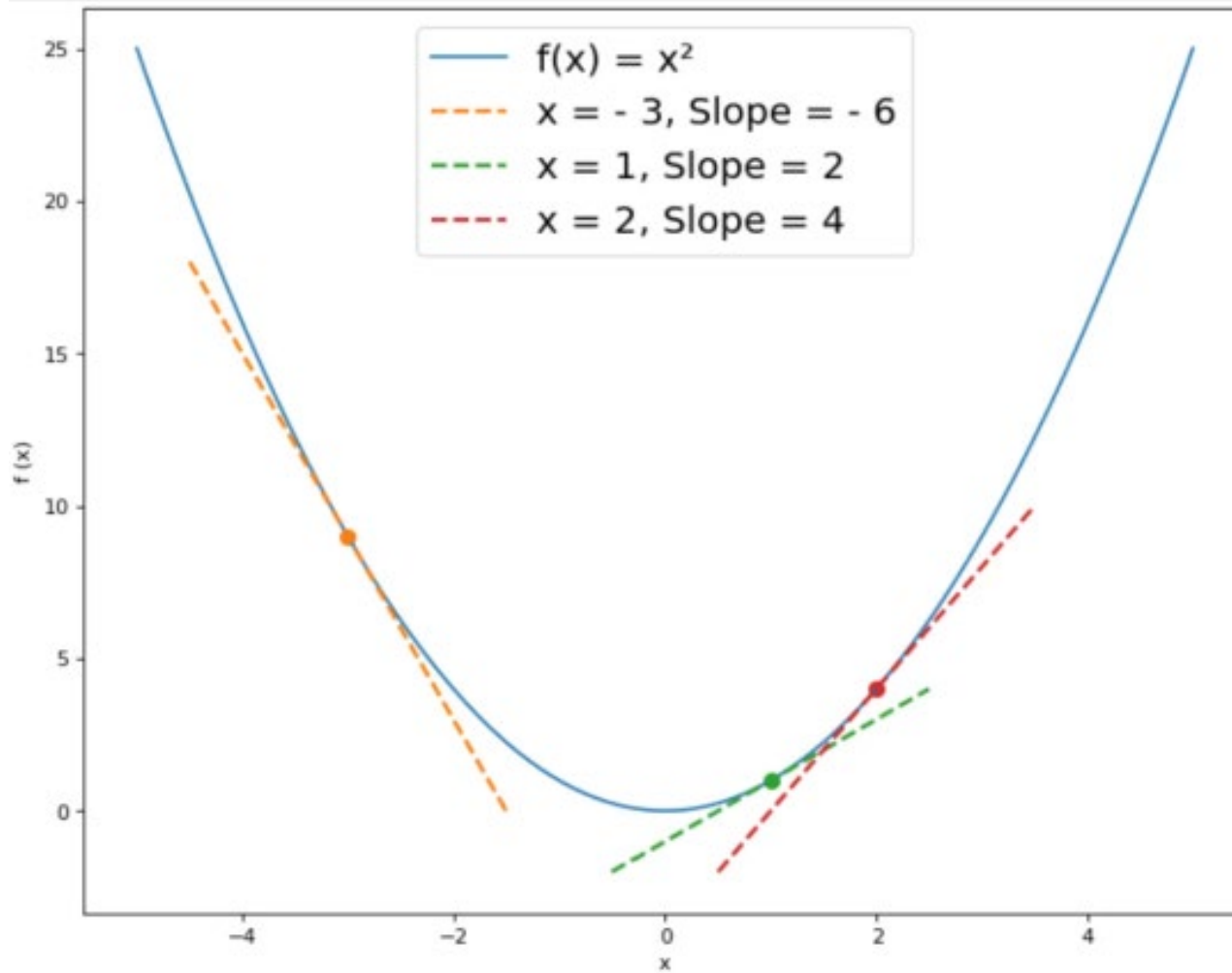
$$GM = \sqrt{\prod_{i=1}^m \frac{\text{right}_m}{N_m}} = \sqrt{\prod_{i=1}^m \frac{N_{ii}}{\sum_{j=1}^m N_{ij}}}$$

$$AMC = \min \frac{\text{right}_m}{N_m} = \min \frac{N_{ii}}{\sum_{j=1}^m N_{ij}}$$

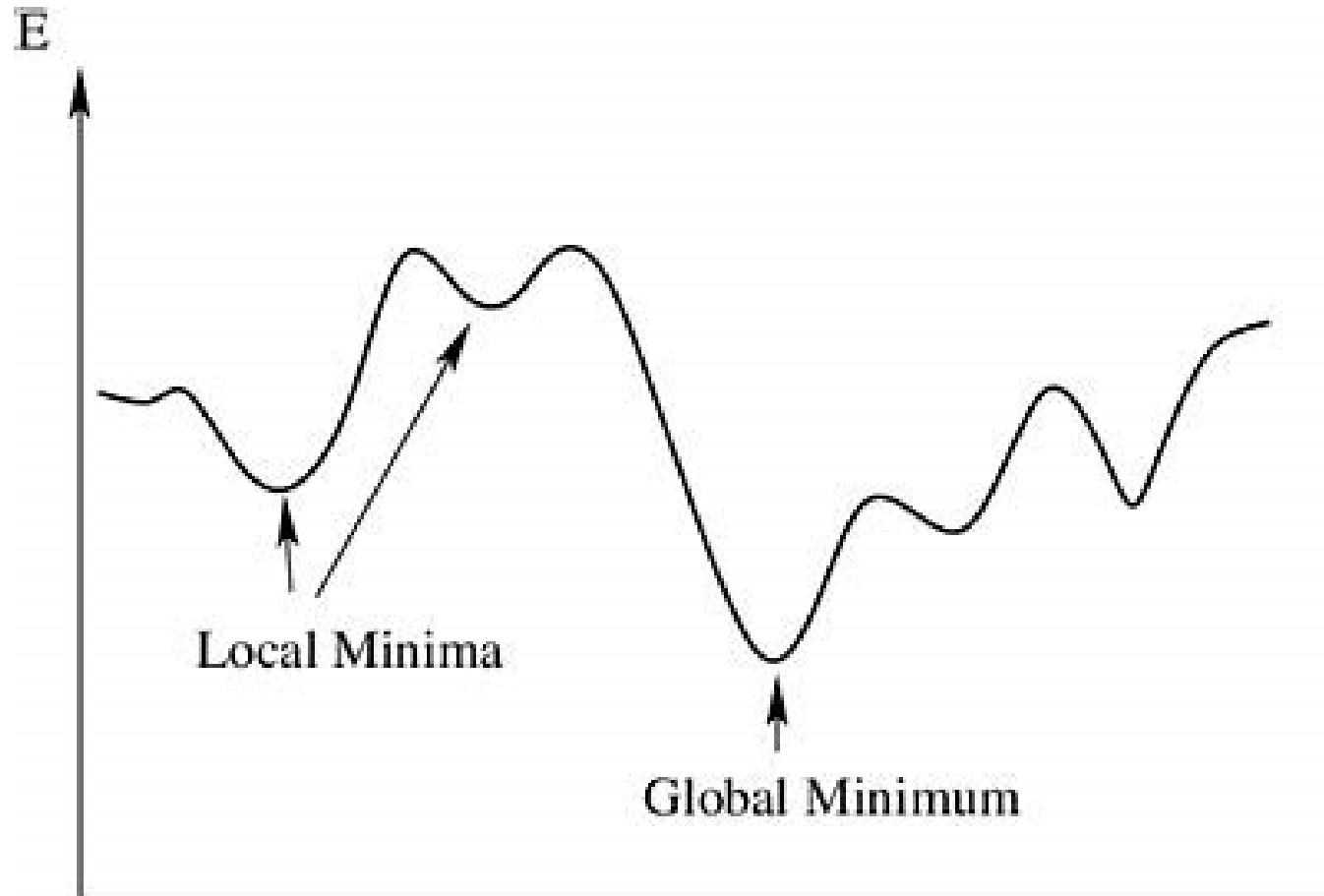
Parameter optimization

Gradient method

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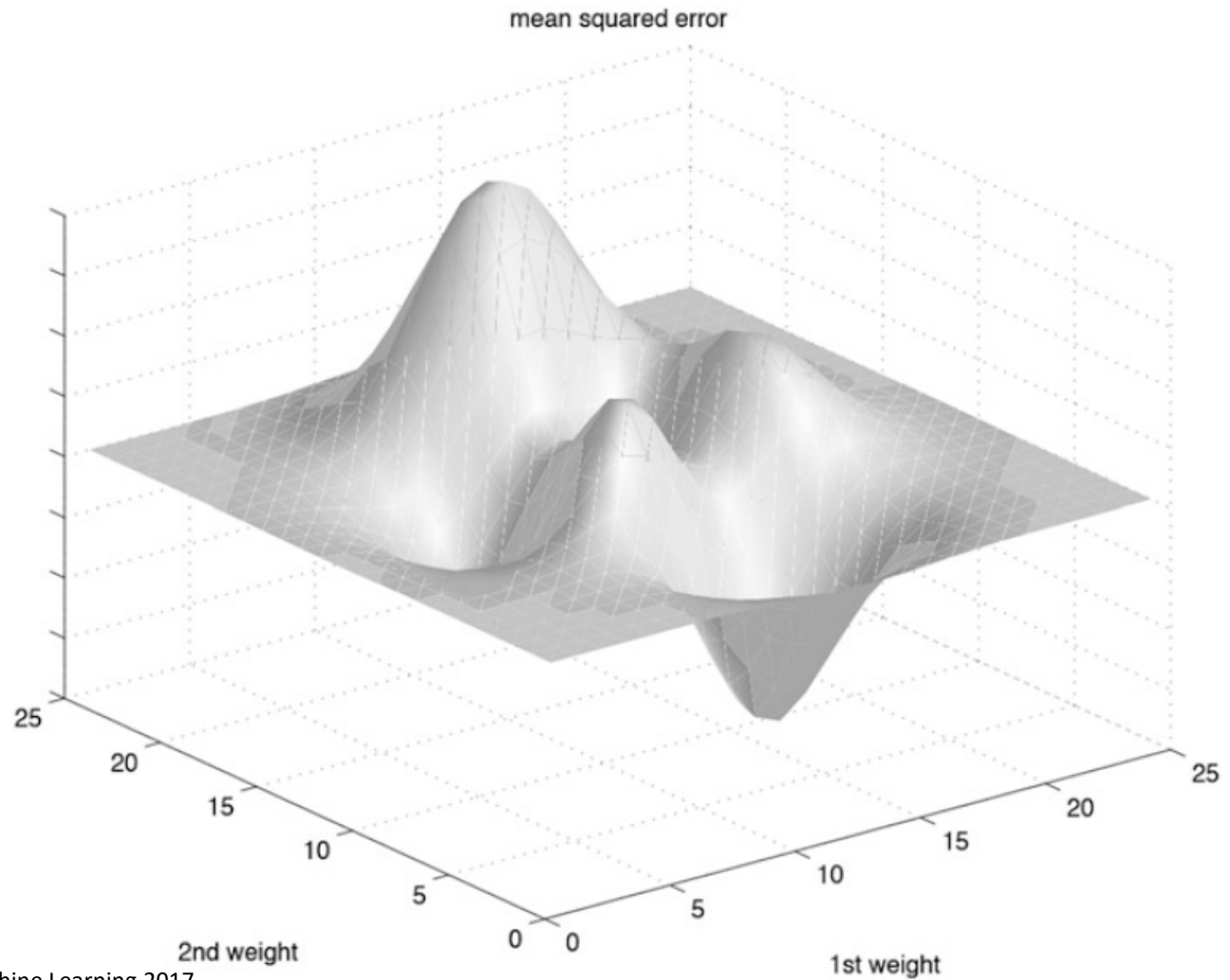
<https://databasecamp.de/ki/backpropagation-grundlagen>



<https://vitalflux.com/local-global-maxima-minima-explained-examples/>

Example Backpropagation

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Forward propagation:

$$y_i = f \left(\sum_j w_{ji}^{(1)} f \left(\sum_k w_{kj}^{(2)} x_k \right) \right)$$

Back propagation:

Output-layer neurons: $\delta_i^{(1)} = y_i(1 - y_i)(t_i - y_i)$

Hidden-layer neurons: $\delta_j^{(2)} = h_j(1 - h_j) \sum_i \delta_i^{(1)} w_{ji}$

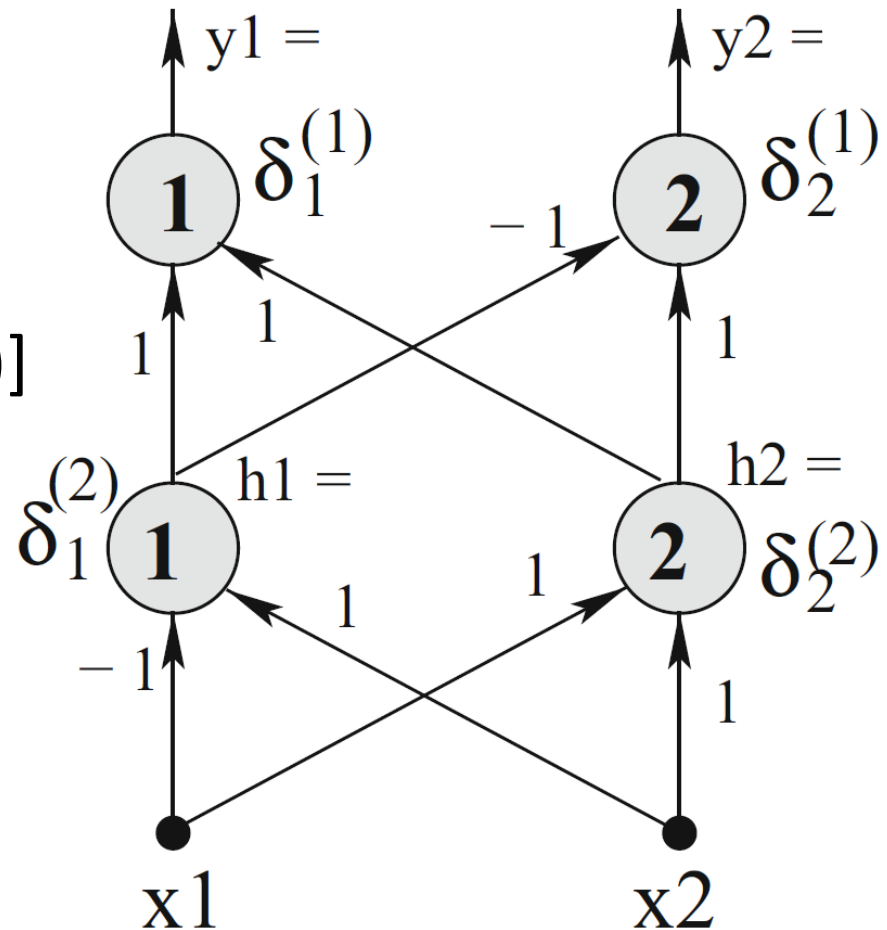
output-layer neurons: $w_{ji}^{(1)} := w_{ji}^{(1)} + \eta \delta_i^{(1)} h_j$

hidden-layer neurons: $w_{kj}^{(2)} := w_{kj}^{(2)} + \eta \delta_j^{(2)} x_k$

Exercise back propagation

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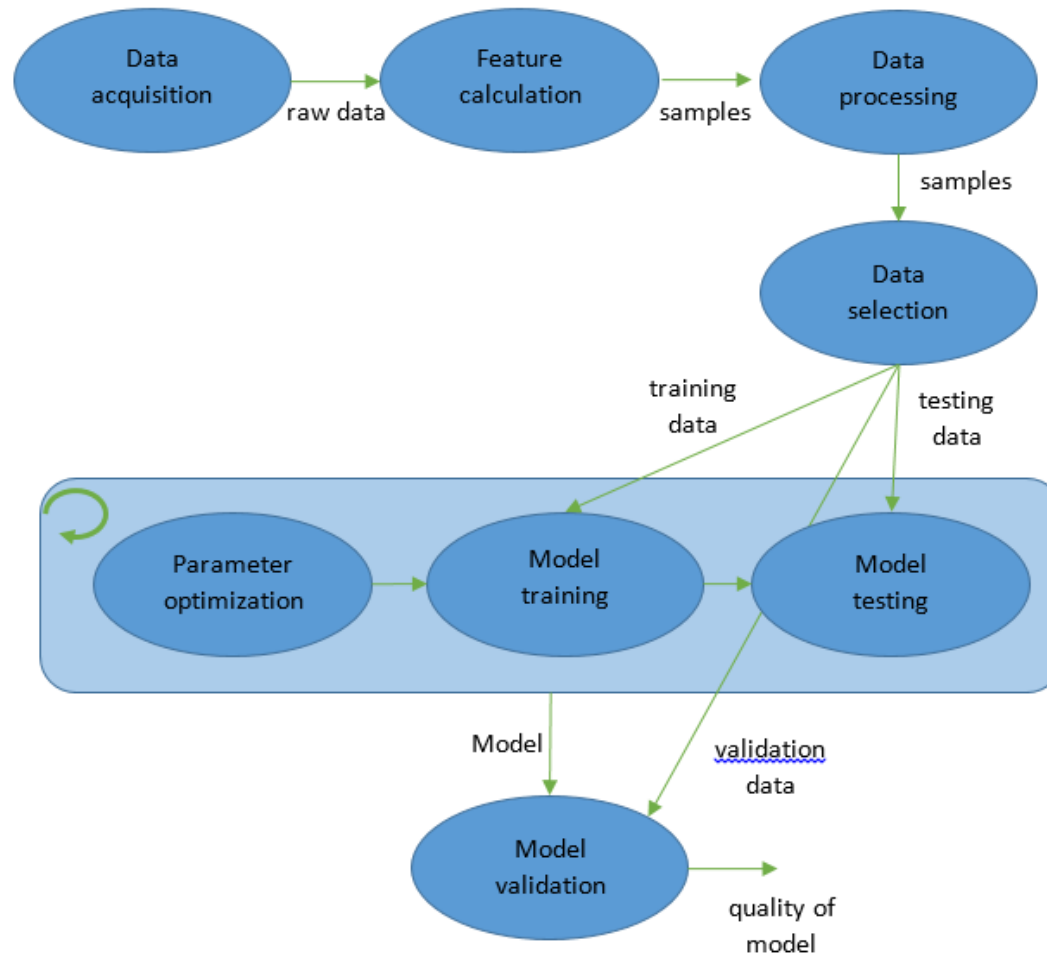
- Transfer function = sigmoid $f(\Sigma) = \frac{1}{1+e^{-\Sigma}}$.
- η - learning rate = 0.1
- input $[\underline{x}, \underline{t}] = [(1, -1), (1, 0)]$



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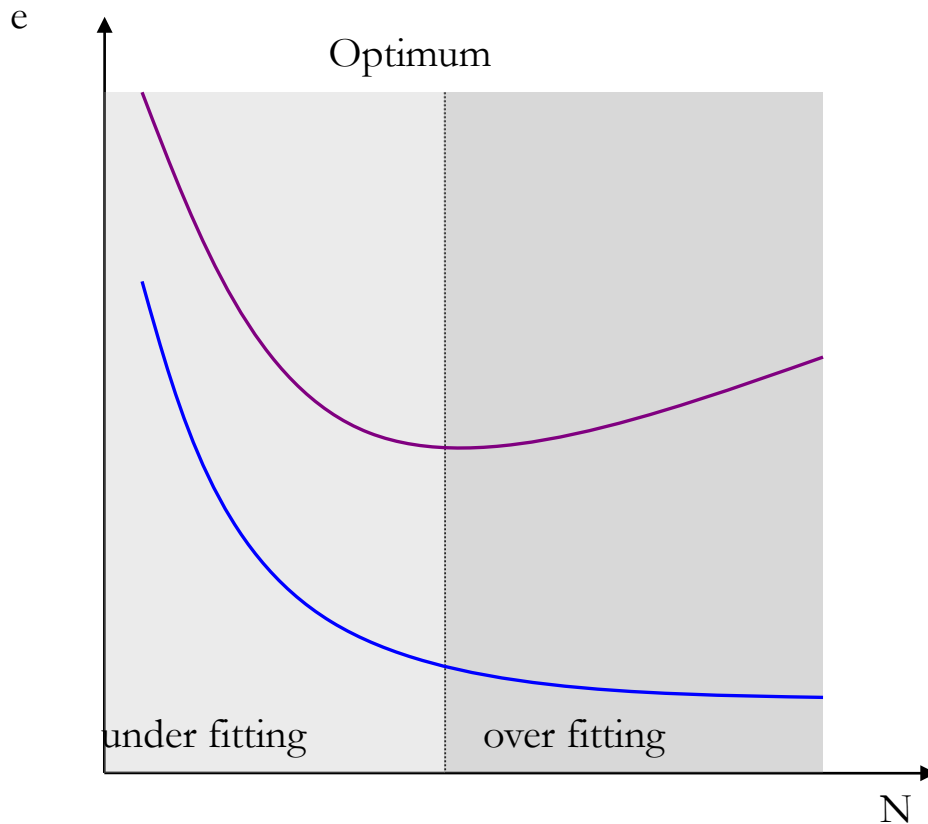
Learning process

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Over- and Underfitting

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e – mean square error

N – number of training cycle

- training data
- validation data
- over fitting
- under fitting
- Optimum

Example Over- and Underfitting with SVM

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